Causal Disentanglement of Location and Semantic Signals in Real Estate Valuation

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Abstract

Automated valuation models increasingly read the prose of a property listing alongside its structured attributes, and the language reliably sharpens their price predictions, yet a gain in predictive accuracy cannot by itself reveal whether the text contributes genuine information about a property or merely relabels the neighborhood geography that an agent's description inevitably carries. We take that ambiguity as the object of study, treating the leading direction of a listing embedding as a high-dimensional treatment whose effect on log sale price we estimate with a partially linear double machine learning score, and we ask how much of that effect survives once location is taken out of the picture. The question is one of spatial confounding, the familiar difficulty in which an exposure and an unmeasured driver of the outcome share a geographic pattern, and the apparatus we assemble around it pairs cross-fitted gradient-boosted nuisances with a closed-form concept-erasure projector that strips latitude and longitude from the representation, a fixed-treatment toggle that adds a flexible location basis to the controls, an intervention that rewrites listings with a large language model, and a finite-sample calibration study. What that apparatus turns up is more reassuring than the predictive-fit worry would lead one to expect. The embedding does encode location, recoverable by a held-out probe at an R^2 reaching one half before erasure and essentially nothing after, and even so, holding the text treatment fixed while adjusting for geography leaves its per-standard-deviation price effect almost unmoved across eleven of twelve metropolitan markets, so that what the language contributes is in most places semantic content rather than geography in disguise. The surviving signal proves robust to the choice of encoder, with a Doc2Vec uniqueness measure and a sentence-BERT principal component tracking one another closely from market to market, while remaining strongly heterogeneous in magnitude and showing its one clear spatial exception in New York, the market whose own text direction is the most geographic of the twelve. Because the estimates are associational rather than cleanly identified, we bound their fragility with a sensitivity analysis that fixes how weak an unobserved confounder would have to be to overturn each market's result, and we document in passing that the influence-function interval double machine learning reports for an embedding-valued treatment undercovers at the sample sizes a single market affords, for which a bootstrap over the full pipeline is the remedy. The contribution is accordingly a disciplined protocol for measuring and deconfounding the price effect of a learned text representation, together with the inferential caution that protocol demands, rather than a causal elasticity of price on language.

Keywords: Spatial Confounding Concept Erasure Double Machine Learning Generated Regressors Text as Data Real Estate Valuation

1 Introduction

Automated valuation models increasingly read the natural-language description of a property alongside its structured attributes, and the text reliably improves their predictions, with Shen and Ross [2021], Baur et al. [2023a], and Lorenz et al. [2023] all reporting sizable reductions in pricing error once listing-text embeddings are added on top of location and the usual structured controls. The reading almost always attached to that improvement is the optimistic one, that listing language carries information about a property which its tabular attributes miss, and on its face the reading is plausible, because a good description does say things about condition, light, and finish that no bedroom count records. What predictive accuracy cannot do, however, is tell that interpretation apart from a more deflationary one, since property descriptions are written by agents who know exactly where a listing sits, so any embedding of those descriptions absorbs the neighbourhood whether the encoder intends it to or not, and a model that prices the embedding may be doing little more than relabelling location as semantic content. Because the two stories make identical predictions, the very gain that motivates text-augmented valuation is by construction silent on whether the language adds anything at all once geography is accounted for, and it is that silence the paper sets out to break.

Breaking that silence means treating the listing embedding as a high-dimensional treatment and asking how much of its association with sale price is owed to geography rather than to anything the language contributes on its own. The question is an instance of spatial confounding, the difficulty in spatial statistics in which an exposure of interest and an unmeasured confounder share a geographic pattern, so that ordinary regression hands the confounder's effect to the exposure [Paciorek, 2010, Hodges and Reich, 2010], and the discipline's usual remedy, restricting the exposure to the orthogonal complement of the spatial effect [Reich et al., 2021, Dupont et al., 2022], has a natural counterpart once the exposure is a learned representation. Whether the representation carries location at all is something we can read off directly, by applying the closed-form concept-erasure projector of Belrose et al. [2023], with an iterative nonlinear pass [Iskander et al., 2023] for the leakage a linear projector leaves behind, and asking how well a held-out probe can still recover the coordinate from what survives. The separate and more consequential question of whether that location actually drives the price effect calls for a separate instrument, and we answer it by holding the text treatment fixed inside a partially linear Double Machine Learning score [Robinson, 1988, Chernozhukov et al., 2018] while moving geography on and off the confounder side, the contrast that a Gelbach decomposition makes precise. Running the framework without exclusion across twelve metropolitan markets then lets the answer be read as a distribution rather than a single number, since the embedding geometry, the structured covariates, and the listing density that decide how much room the text has once location is accounted for all differ from one market to the next.

The deconfounding test that this design supports returns an answer more reassuring than the predictive-fit worry would suggest, and the route it takes to that answer is worth following carefully, because the most obvious reading of the erasure diagnostic seems at first to point the other way. A held-out probe recovers the latitude-longitude coordinate from the embedding at an R^2 that climbs toward one half, so the representation unmistakably carries location, and the natural expectation is that the price effect of its leading direction would dissolve once that geography were taken out, and yet that is not what happens, because holding the text direction fixed as the treatment and folding a flexible location basis into the controls leaves its per-standard-deviation effect on log price almost exactly where it began across eleven of the twelve markets, while the single market in which the adjustment does bite, New York, turns out to be the one whose text direction is itself the most geographic, so that the exception illuminates the mechanism rather than embarrassing it. What the design settles about the magnitude of the effect is matched by what it forces us to confront about the standard error, since the leading principal component is estimated on the very sample that estimates the price effect and therefore behaves as a generated regressor in the sense of Pagan [1984], the orthogonal-score variance of Chernozhukov et al. [2018] does not absorb the extra sampling variability this introduces, and the analytic interval comes out anticonservative at the few thousand listings a single metropolitan market affords, a shortfall we document in a calibration study and repair with a bootstrap that resamples the entire pipeline. Beneath the generated-regressor problem sits the eigengap of the embedding covariance, and a sin-Θ stability argument ties how faithfully the estimated treatment direction tracks its population target to the size of that gap, which is finally what licenses treating the leading component as a stable object to put on the left of a causal question.

Applied across twelve metropolitan areas using 3,851 deduplicated Redfin listings, the framework returns a picture whose parts hang together. Listing text carries a positive per-standard-deviation association with log sale price that holds up regardless of which encoder produces it, since a Doc2Vec uniqueness channel and a sentence-BERT principal component move almost in lockstep from market to market and so read as two encodings of a single signal rather than as independent confirmations of it. That association varies strongly in magnitude across the twelve markets, it survives the spatial deconfounding just described everywhere but New York, and it is not organised by any metropolitan demographic moderator that withstands multiplicity adjustment and leave-one-out leverage diagnostics on so few units. None of this amounts to a clean causal elasticity, and we are careful to say so throughout, because the estimates remain associational rather than identified by design, and the sensitivity analysis we run in the style of Cinelli and Hazlett [2020] fixes how weak an unmeasured confounder would have to be to overturn each market's result, a threshold that comes out uncomfortably modest in several of them. What the paper offers in the end is a disciplined protocol for measuring and deconfounding the price effect of a learned text representation, carried out with inference calibrated to the embedding-treatment setting, and we release the data and the code so that the protocol can be reused rather than only read.

We have arranged what follows in the order a skeptical reader would want to work through it. The literatures on spatial confounding, on concept erasure, and on text as a treatment have grown up largely apart from one another, and Section 2 brings them into contact, after which the estimand and the linear and nonlinear erasure pipeline on which the later analysis leans are developed in Section 3. How the corpus was assembled, and audited for the duplication that scraped listings invite, is the business of Section 4, and the calibration study behind our inferential caution is documented in Appendix I, where the influence-function interval is shown to undercover for an embedding treatment and an out-of-fold construction of the treatment direction restores most of the lost coverage. The twelve-market evidence occupies Section 5, which moves in turn through the uniqueness and principal-component estimates, the rewriting intervention, the erasure and decomposition diagnostics, the sensitivity bounds, and the cross-method comparison, before Section 6 closes by drawing the line, for practice, between what a predictive-only evaluation of text-augmented valuation can and cannot support.

2 Related Work

2.1 Listing text in automated valuation, and the question it raises

The use of natural language in property valuation predates the recent enthusiasm for embeddings, and the empirical claim that listing text improves prediction is by now well established. Nowak and Smith [2017] show that the free-text remarks attached to a multiple-listing record reduce pricing error substantially once they are encoded, and Nowak et al. [2021] document that the vocabulary doing the predicting shifts from one submarket to the next. A particular channel is formalised by Shen and Ross [2021], who measure the textual uniqueness of a description relative to its neighbours and report a uniqueness premium on the order of ten to fifteen percent per standard deviation in a single metropolitan market, while Baur et al. [2023a] show that folding a sentence-embedding representation into a structured valuation model lowers its error by as much as a sixth. With few exceptions the literature reads gains of this kind as evidence that language carries information about a property which its tabular attributes do not.

What that reading passes over is that an agent writes a description already knowing exactly where the property sits, so the language is saturated with location before any encoder touches it, and a representation learned from such text will carry the neighbourhood whether or not it carries anything more. Because a predictive metric cannot tell a model that prices genuine semantic content apart from one that prices laundered geography, the very gains that motivate text-augmented valuation are silent on which of the two is operating, and the question of how much of the apparent text effect is real has gone largely untested in a causal register even as the predictive work has matured. We take up that question directly, and the apparatus it calls for sits at the meeting point of three literatures that have seldom been brought into the same room, namely the treatment of text as a causal variable, the spatial-statistics account of confounding by location, and the concept-erasure methods developed in natural language processing for removing a named attribute from a representation. The same machinery bears on the fairness and auditability concerns that motivate much of the recent scrutiny of automated valuation, since Neal et al. [2024] and Wang et al. [2025] document residual valuation gaps that survive architectural improvement and a text feature re-encoding neighbourhood geography re-encodes whatever that geography is correlated with, but the object of this paper is the prior measurement question of whether the text effect is genuine at all rather than the disparities a confirmed effect would carry.

2.2 From Hedonic Models to Multimodal Valuation

The idea that property values can be decomposed into contributions from observable characteristics dates to the hedonic pricing framework [Rosen, 1974, Sheppard, 1999]. Automated valuation models built on this foundation through progressively richer statistical machinery: generalized additive models [Gloudemans, 1999], spatial autoregressive specifications that account for price correlation among neighboring properties [Pace and Barry, 1998, Can, 1992], and kriging-based approaches that interpolate unobserved spatial surfaces [Bourassa et al., 2010]. Each generation recognized more clearly that spatial dependence is not a nuisance to be corrected but a first-order feature of housing markets.

Machine learning accelerated this trajectory. Ensemble methods outperform linear hedonic models by capturing non-linear feature interactions automatically [Park and Bae, 2015, Antipov and Pokryshevskaya, 2012], and deep learning extended the input space beyond tabular features to images [Poursaeed et al., 2018, Law et al., 2019a, You and Liu, 2017] and text [Shen and Ross, 2021, Baur et al., 2023a]. Visual models trained on property photographs or street-view imagery can estimate luxury level and architectural style [Poursaeed et al., 2018, Lindenthal and Johnson, 2021], while textual models extract semantic features from listing descriptions that correlate with sale prices [Nowak and Smith, 2017, Law et al., 2019b]. Multimodal systems fuse these modalities together, reporting performance gains attributed to complementary information [Bin et al., 2020, Kang et al., 2021, Kostić and Jevremović, 2020]. Graph neural networks push the idea further by propagating information between spatially proximate properties, so that each node's representation absorbs its neighbors' features [Das et al., 2021, Lin et al., 2021, Peng et al., 2022, Li et al., 2018, Huang et al., 2018].

A persistent concern runs through this literature, raised early but seldom addressed. The most predictive words in property descriptions vary across geographic submarkets [Nowak and Smith, 2017, Nowak et al., 2021], visual features capture neighborhood-level rather than property-level variation [Law et al., 2019a], architectural style clusters geographically [Lindenthal and Johnson, 2021], and GNN message passing, by design, propagates whatever signal is shared among neighbors, including location-induced price similarity. If text, images, and graph features all encode geography, then fusing them creates the appearance of complementary signal from what is really redundant spatial encoding through different channels.

2.3 The Semantic Interpretation and Its Limits

The commercial and research appeal of text-based valuation rests on a specific claim: that property descriptions carry information about value that structured features miss. A one-standard-deviation increase in textual "uniqueness" is reported to raise predicted price by 10–15% [Shen and Ross, 2021]. Including text embeddings is reported to reduce valuation error by up to 17% [Baur et al., 2023a]. Hierarchical graph representations that embed similar-value properties near one another in latent space appear to learn price-relevant structure [Zhang et al., 2021]. UMAP projections of these embeddings produce visually appealing clusters that seem to correspond to market segments [Baur et al., 2023a].

Every one of these findings is also predicted by a model in which text carries zero independent information. If agents in affluent neighborhoods write "luxury finishes" and "steps from fine dining" while agents in family-oriented suburbs write "great school district" and "quiet cul-de-sac," the embeddings will cluster by price segment, SHAP will rank descriptive terms as important [Lorenz et al., 2023, Stang et al., 2023], and adding text to a location-only model will improve R^2, even though the entire gain comes from spatially confounded language rather than property-specific content. The assumption that generic quality terms like "spacious" or "renovated" provide location-independent signal [Lorenz et al., 2023] ignores that "spacious" is more frequent in suburban markets and "renovated" more frequent in gentrifying urban neighborhoods. Nowak et al. [2021] find exactly this: the predictive vocabulary shifts across submarkets, suggesting that language mirrors geography rather than transcending it. Wan and Lindenthal [2023] raise the same concern for visual features, proposing system-level tests to determine whether models respond to building attributes or to irrelevant context.

The distinction between correlation and causation is not academic here. If text embeddings genuinely encode value-relevant property attributes independent of location, then NLP-enhanced valuation models offer real information gains. If they primarily encode geography through linguistic patterns, then these models are opaque spatial models and their accuracy gains are expected, not evidence of new signal. Yet across the studies reviewed above, none tests the causal claim. The tools to do so exist in causal inference and spatial statistics, and the confounding structure is transparent, but the question has not been asked. This paper asks it.

2.4 Spatial Confounding and Attribute Leakage in Learned Representations

The concern that text may encode geography rather than semantics connects two literatures that have developed largely in parallel.

In spatial statistics and epidemiology, the problem is well characterized: when an exposure of interest varies spatially and shares its spatial pattern with unmeasured confounders, standard regression attributes the confounders' effects to the exposure [Paciorek, 2010]. The severity depends on the spatial scale: fine-grained exposures are most vulnerable to fine-grained confounders. Methods to address this include restricted spatial regression, which orthogonalizes spatial random effects against fixed effects of interest [Reich et al., 2021], and sensitivity frameworks that quantify the confounder strength needed to overturn a finding [Cinelli and Hazlett, 2020]. A counterintuitive result from this literature is that adding spatially-correlated error terms to a model can worsen, not improve, inference about fixed effects when spatial confounding is present [Hodges and Reich, 2010]. Spatial econometrics addresses the same challenge through spatial lag and error models [Anselin, 1988, LeSage and Pace, 2009, Elhorst, 2014], though identification of spatial dependence parameters remains difficult to separate from omitted-variable bias [Gibbons and Overman, 2014].

In NLP, a parallel line of work has shown that text embeddings absorb far more than semantic content from their training corpora. Word embeddings encode human-like gender and racial biases [Bolukbasi et al., 2016], replicate implicit association test results [Caliskan et al., 2017], and retain these biases even after standard debiasing [Gonen and Goldberg, 2019]. Language use varies systematically by geography: user location can be predicted from tweets with high accuracy [Rahimi et al., 2018], and lexical innovations diffuse through social networks along spatial paths [Eisenstein et al., 2014]. Property descriptions are an extreme case of this phenomenon because agents write with complete knowledge of the property's location, neighborhood, and market context. The language is not incidentally spatial; it is intentionally so.

Machine learning has begun to develop tools for this class of problem. Causal representation learning seeks representations that reflect causal structure rather than spurious correlation [Schölkopf et al., 2021, Peters et al., 2017]. Counterfactual fairness formalizes the requirement that predictions be invariant to interventions on protected attributes [Kusner et al., 2017, Kilbertus et al., 2017]. Domain-adversarial networks learn confounder-invariant representations through gradient reversal [Ganin et al., 2016], and variational approaches attempt to disentangle causal from spurious factors [Louizos et al., 2017, Moyer et al., 2018]. The invariance of causal mechanisms across environments, in contrast to the instability of correlations, motivates transfer-based identification [Rojas-Carulla et al., 2018]. Causal discovery algorithms infer graph structure from data using conditional independence tests [Spirtes et al., 2000], score-based search [Chickering, 2002], or functional asymmetries between causal directions [Shimizu et al., 2006, Peters et al., 2016]. Double Machine Learning [Chernozhukov et al., 2018] provides a semiparametric framework for estimating causal effects with machine-learned nuisance parameters at $\sqrt{n}$ rates.

These threads meet in the setting we study, and they have begun to meet elsewhere as well, which is worth stating precisely so that what is new here is not overstated. The idea of adapting a text representation for causal rather than predictive ends is developed by Veitch et al. [2020], whose embeddings are built to retain the confounding-relevant subspace, and the inverse operation of stripping a concept from a document embedding by linear erasure is carried out for similarity and clustering by Fan et al. [2025], while Feldman et al. [2026] treat latent textual features as a treatment whose covariate content has to be residualised away. What this paper adds is therefore neither the erasure operator nor the recognition that a text feature can conflate treatment with confounder, both of which are by now established, but the application of that apparatus to an economic estimand, the price effect of a listing's language, carried out with a design that separates the geography a representation merely encodes from the geography that actually moves the outcome and with inference suited to an embedding-valued treatment. Property valuation is an unusually clean place to attempt this, since location drives both the text and the price, the outcome is measured precisely, and the practical stakes are immediate, and we run the analysis across twelve metropolitan markets rather than the single market the hedonic-text literature has usually examined, which lets the answer be read as a distribution and lets the one market where location does bite be identified rather than assumed away.

3 Methodology

3.1 Setup and the conflation it invites

Write $\mathcal{D} = \{(x_i, t_i, \ell_i, y_i)\}_{i=1}^n$ for a market of n listings, where $x_i \in \mathbb{R}^d$ collects the structured property attributes that hedonic models have always used, the bedroom and bathroom counts, the living area, the lot size, and the year built; $\ell_i \in \mathbb{R}^2$ is the latitude-longitude coordinate of the parcel; y_i is the log sale price; and t_i is a representation of the listing's free-text description, either a scalar summary or a high-dimensional embedding depending on the channel. A text-augmented valuation model learns some $\hat{y}_i = f(x_i, t_i, \ell_i)$, and the empirical regularity that motivates this literature is that adding t_i to the feature set raises predictive accuracy beyond what x_i and ℓ_i deliver on their own.

The difficulty is that an accuracy gain cannot say what kind of information the text carried. An agent who writes "sun-drenched and steps from the park" is describing a property, but the same phrase is also a near-deterministic signal of which block the property sits on, because language and location are produced together by an author who observes both. A predictive model that prices genuine semantic content and a predictive model that prices laundered geography make identical out-of-sample predictions, so the gain on its own leaves an analyst unable to tell which of the two has been bought. Our object of interest is the part of the text-price relationship that is not the spatial information the text carries, and the methods that follow are built to isolate it and then to ask, for each metropolitan market in turn, how much of the apparent text signal that isolation leaves standing.

3.2 A structural account of the confounding

The data-generating process we have in mind makes the confounding explicit rather than assuming it away. Location is exogenous and drives everything downstream: the structured attributes x, a vector c of contextual neighborhood features such as demographics, amenities, and recent crime, the text t, and the price y. The text in turn is written with knowledge of location, structure, and context, so $t = f_t(\ell, x, c, U_t)$, and the question the paper exists to answer is whether t enters the structural equation for y at all once ℓ, x, and c are accounted for, or whether the entire observed association between text and price runs through the common dependence of both on location.

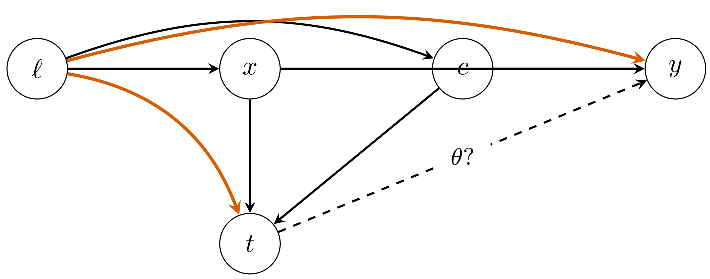

Figure 1: The structure the paper adjudicates. Location ℓ is exogenous and feeds the property attributes x, the contextual features c, the listing text t, and the price y. The two highlighted edges $\ell \to t$ and $\ell \to y$ form the spatial-confounding backdoor, the reason text and price are associated in part simply because both depend on where the property sits. The dashed edge labelled θ is the direct effect of text on price, the quantity we estimate, and the empirical question is how much of the text-price association is that direct edge rather than the backdoor through ℓ. Conditioning on $\{\ell, x, c\}$ blocks the backdoor and identifies θ; whether θ is zero or positive is what the twelve markets decide.

Two readings of this diagram bracket the possibilities. Under one, the text effect on price is zero net of location and the structured covariates, every observed text-price correlation is geography wearing a semantic costume, and the apparent predictive value of listing language is spatial confounding mediated through words. Under the other, listing text carries property-specific content that the structured covariates miss, so t does enter the equation for y and the text-price relationship is at least partly a genuine semantic effect. The two readings disagree about a quantity that is identified once the backdoor set $\{\ell, x, c\}$ is conditioned on, since that set blocks every path from t to y that does not pass through the hypothesized direct edge, and the rest of this section develops the estimator, the encodings, and the deconfounding tests through which the twelve markets adjudicate between them.

3.3 The partially-linear estimand and its estimation

We summarize the listing text by a scalar treatment T, defer to Section 3.4 how that scalar is constructed, and pose the partially-linear model

$$Y = \theta T + g(\ell, x, c) + \varepsilon, \qquad T = m(\ell, x, c) + \nu, \tag{1}$$

in which θ is the per-standard-deviation effect of the text direction on log price and the nuisance functions g and m are left unrestricted. The Robinson partialling-out estimator [Robinson, 1988] recovers θ from the residual-on-residual regression of $\tilde{Y} = Y - \hat{g}$ on $\tilde{T} = T - \hat{m}$, and we fit $\hat{g}$ and $\hat{m}$ by cross-fitted gradient boosting in the double machine learning framework of Chernozhukov et al. [2018], splitting each market into $K = 5$ folds and predicting each fold's residuals from models trained on the complement so that the nuisance estimates and the score are never formed on the same observations. The resulting estimator is Neyman-orthogonal and root-n consistent whenever the nuisance fits converge faster than $n^{-1/4}$, and its influence-function variance gives the standard error we report by default,

$$\hat{\theta} = \frac{\sum_i \tilde{T}_i \tilde{Y}_i}{\sum_i \tilde{T}_i^2}, \qquad \hat{\mathrm{SE}}_\theta = \sqrt{\frac{1}{n^2} \sum_i \frac{(\tilde{Y}_i - \hat{\theta}\, \tilde{T}_i)^2\, \tilde{T}_i^2}{\left(\frac{1}{n} \sum_j \tilde{T}_j^2\right)^2}}. \tag{2}$$

One feature of the embedding channel complicates the inference and we flag it here because it shapes how the per-market intervals should be read. When the treatment is the leading principal component of the embedding, that direction is itself estimated on the same sample that estimates the price effect, so T is a generated regressor in the sense of Pagan [1984] and the orthogonal-score variance in Equation (2) does not absorb the extra sampling variability the estimation of the direction introduces. The interval is correspondingly anticonservative at the few thousand listings a single market affords, mildly so near the null and more seriously when the true effect is large, a pattern we document in a calibration study and repair with an out-of-fold construction of the treatment direction in Appendix I. The point estimates are approximately unbiased throughout; it is the nominal coverage of the analytic interval, not the location of the estimate, that the calibration study qualifies.

3.4 Two encodings of the text treatment

Because no single reduction of a paragraph to a number is obviously the right one, we read the listing text through two encodings with different inductive biases and treat their agreement as a robustness check rather than as independent corroboration. The first follows the hedonic specification of Shen and Ross [2021], who summarize a description by a Doc2Vec uniqueness score that measures how far a listing's language departs from the corpus norm, and we enter that scalar into Equation (1) as the treatment. The second follows Baur et al. [2023a], who embed each description with a pretrained sentence transformer and read the leading principal component of the embedding as the text signal; we standardize that component and use it as the treatment in a separate partialling-out fit. The sign of a principal component is not identified, since a component and its negation span the same axis, so wherever the Baur channel appears we orient the common axis to make the pooled estimate non-negative and disclose the orientation as a convention rather than a finding, a convention that cannot manufacture agreement because the cross-market correlation between channels is invariant in magnitude under a sign flip of either one.

3.5 What the embedding encodes: erasure as a diagnostic

The sentence-transformer embedding that underlies the Baur channel was pretrained on generic text and has no reason to keep semantic content about a property separate from the geographic information that correlates with its neighborhood, so before asking whether the price effect is confounded we ask the prior question of how much location the representation carries at all. We answer it by erasing the coordinate concept from the embedding and measuring what an erasure leaves recoverable. The closed-form concept erasure of Belrose et al. [2023] removes the linear subspace that the mean-conditional-covariance estimator associates with the latitude-longitude pair and comes with an exact-guardedness guarantee that no linear predictor can recover the coordinate from the residual better than a constant baseline, and the iterative nonlinear procedure of Iskander et al. [2023] extends the attack to the directions a linear probe cannot see by alternating between fitting a nonlinear probe of the coordinate and projecting out where that probe concentrates its predictive mass. The erasure is a diagnostic of what the representation contains and not a modification of any per-market estimate, since the Baur effects are computed on the unprojected embedding in keeping with the published protocol; its role is to bound how much location a linear or nonlinear function of the text could carry, which the decomposition below then connects to how much location actually moves the price.

3.6 When a probe fails, what has been erased?

The erasure diagnostic of the previous subsection rests on probing, and probing carries a subtlety that bears on how its verdicts should be read. The natural way to check that a concept has left a representation is to train a probe of the concept on the cleaned representation and confirm that it does no better than chance, yet a probe that fails establishes only that this particular probe, at this capacity, cannot recover the concept, and not that the representation has shed it, since a fresh classifier of higher capacity fit after the fact can recover what a weaker or adversarially pressured probe could not. This is the reason we lean on closed-form erasure, whose guardedness guarantee holds against every linear probe at once rather than against the one we happened to train, and the reason we read the residual probe R^2 that survives the iterative erasure of Section 5.3 as a lower bound on the location a nonlinear function could still recover rather than as a measurement of zero. The same subtlety is a warning about the obvious alternative to closed-form erasure, an adversarially trained encoder whose discriminator gradient reversal drives to chance, because a live discriminator at chance is the weakest possible evidence of erasure: it certifies only that the encoder defeated the one probe it was trained against, and a fresh post-hoc probe of greater capacity may still read the concept off the same representation. We make the point precise as a $\mathcal{V}$-information [Xu et al., 2020] statement connected to the Barber–Agakov variational lower bound used throughout the probing literature [Pimentel et al., 2020].

For any class $\mathcal{F}$ of probabilistic classifiers $q : \mathcal{Z} \to [0, 1]$, define the variational lower bound on $I(Z; C)$:

$$V_{\mathcal{F}}(Z; C) \;:=\; \sup_{q \in \mathcal{F}} \big[H(C) - \mathbb{E}_{(Z,C)}[-\log q(C \mid Z)]\big]. \tag{3}$$

The supremum is achieved at $q^\star = p(C \mid Z)$ when $\mathcal{F}$ contains the Bayes-optimal posterior; otherwise $V_{\mathcal{F}} < I(Z; C)$ strictly.

Proposition 1 (Frozen-probe diagnostic)**.** *Let $E_\phi : \mathcal{X} \to \mathcal{Z}$ be an encoder, Φ the discriminator class trained jointly via gradient reversal, and Φ' a probe class with $\Phi \subseteq \Phi'$ trained post-hoc on the frozen encoder. Then:*

(a) $0 \le V_\Phi(Z; C) \le V_{\Phi'}(Z; C) \le I(Z; C)$, *with the rightmost inequality strict whenever Φ' omits the Bayes-optimal posterior.*

(b) The empirical Barber–Agakov lower bound under Φ equals the gradient-reversal discriminator's converged value: $\hat{V}_\Phi(Z_{\hat\phi}; C) = H(C) - \hat{L}_\Phi$. Therefore "the live discriminator achieves chance" is equivalent to $\hat{V}_\Phi \le \epsilon$. Under uniform-convergence conditions on Φ', the post-hoc plug-in $\hat{V}^{post}_{\Phi'}(Z_{\hat\phi}; C) \to V_{\Phi'}(Z_{\hat\phi}; C)$ in probability, and hence $\hat{V}^{post}_{\Phi'} - \hat{V}_\Phi$ is a consistent lower-bound estimator of the residual mutual information $I(Z_{\hat\phi}; C) - V_\Phi(Z_{\hat\phi}; C) \ge 0$.

(c) There exist a data distribution over (X, C), a downstream label Y with $I(X; Y) > 0$, class pairs $\Phi \subsetneq \Phi'$, and a joint-game weight $\alpha > 0$ such that the joint adversarial-prediction loss

$$\mathcal{L}(\phi, g, h) \;=\; \alpha \cdot \mathcal{L}_{task}(g(E_\phi(X)), Y) \;-\; \mathcal{L}_{disc}(h(E_\phi(X)), C)$$

admits a saddle $(\phi^\star, g^\star, h^\star)$ with $V_\Phi(Z_{\phi^\star}; C) = 0$ but $V_{\Phi'}(Z_{\phi^\star}; C) \ge H(C) - \delta$ for arbitrary $\delta > 0$.

Statement (a) is folklore in the variational mutual-information literature, appearing as Proposition 1 of Xu et al. [2020] in the language of $\mathcal{V}$-information and as the central observation of Pimentel et al. [2020] in the language of probe capacity, and we restate it for self-containedness. Statements (b) and (c) are our contribution. Part (b) ties the bound to the gradient-reversal objective and recasts the post-hoc adversary used as a heuristic by Moyer et al. [2018] and Elazar and Goldberg [2018] as a consistent statistical estimator of residual leakage, which is what licenses reading the surviving probe R^2 as a number rather than a guess. Part (c) establishes that the joint adversarial-prediction game admits saddles at which the live discriminator achieves the optimum of its own minimax game while essentially all of the concept remains recoverable to a richer probe, so an adversarial erasure can certify success and be wrong. The existential construction mirrors the style of Ravfogel et al. [2022] Proposition 3.2 and Belrose et al. [2023] Theorem 4.3, and proofs together with three synthetic numerical experiments verifying each part are deferred to Appendix G.

3.7 Holding the text fixed: a covariate decomposition

Showing that the embedding encodes geography is not the same as showing that the price effect is confounded by it, because a representation can carry a confounder along directions that have nothing to do with the one on which the treatment acts. The test that speaks to confounding directly holds the text treatment fixed and moves geography across the conditioning set, in the manner of the order-invariant covariate decomposition of Gelbach [2016]. We estimate the text effect twice on the same fixed treatment, once with the location coordinates held out of the controls and once with a flexible spatial basis, the coordinates together with their squares and their interaction, folded in, and read the difference between the two estimates as the share of the text effect that explicit geography accounts for. A small difference means the text-price effect is not the spatial information re-expressed, and the decomposition is informative rather than mechanical precisely when the text direction is shown by the erasure diagnostic to carry geography, since a direction that encodes location yet whose price effect is unmoved by conditioning on location is the signature of a genuine semantic signal. This contrast is the paper's headline deconfounding test, and Section 5.4 reports it for every market.

3.8 An intervention check and a sensitivity bound

Two further instruments probe the same question from angles with different failure modes. A counterfactual rewriting intervention uses an open-weight language model to edit each description under a minimal-pair protocol, producing one rewrite that strips stylistic content while holding the model's inferred submarket fixed and a second that also moves the targeted submarket, and the price differentials these edits imply isolate a within-listing style contrast and a location-inference contrast whose biases are not shared with the cross-listing geometry of a fixed encoder. Because none of these designs identifies a structural effect by assumption, we close with the sensitivity calculus of Cinelli and Hazlett [2020], reporting for each market the robustness value, the strength an unobserved confounder would need on both the treatment and the outcome to overturn the estimate, benchmarked against the strength of the spatial covariates we do observe. The twelve per-market estimates are finally combined through a random-effects meta-analysis with the Paule-Mandel variance estimator and Hartung-Knapp-Sidik-Jonkman small-sample intervals [Paule and Mandel, 1982, IntHout et al., 2014], and a moderator scan asks, with multiplicity and leverage diagnostics, whether any metropolitan covariate explains the heterogeneity the markets display.

4 Dataset Construction

4.1 A twelve-market corpus of listings

The corpus is built from sold residential listings scraped from a major online real estate platform across twelve metropolitan markets that span the price and geographic range of the United States housing stock, from San Francisco and New York at the expensive coastal end through Boston, Seattle, Washington, Denver, and Portland in the middle to Chicago, Atlanta, Phoenix, Dallas, and Philadelphia toward the more affordable interior and south. Each listing carries the free-text description an agent wrote, the realized sale price the platform records, the structured attributes a hedonic model expects, the bedroom and bathroom counts, the living area, the lot size, the year built, and the property type, and a property-level latitude and longitude

Table 1: The twelve-market corpus. *Listings* is the per-market analysis sample after deduplication to one row per listing and removal of rows missing a required field; the markets together contribute 69,173 listings. Median sale price and median description length are computed on the analysis sample. The panel ranges over more than an order of magnitude in size and roughly fivefold in price.

Market	Listings	Median price	Median words
New York	15,227	$975,000	183
Dallas	8,008	$439,000	161
Phoenix	7,086	$514,990	117
Philadelphia	7,030	$255,000	134
Atlanta	6,134	$384,000	178
Chicago	5,576	$374,900	129
Denver	5,280	$525,000	207
Portland	4,337	$559,000	166
Washington	4,015	$599,000	187
Seattle	2,884	$750,000	152
Boston	2,610	$999,000	137
San Francisco	986	$1,295,000	174
Total	69,173		

rather than a coarse centroid. Working with recorded sale prices rather than the assessed values or tax-reassessment proxies that earlier versions of this project relied on removes a layer of measurement concern at the root, since the outcome is the transaction the market actually cleared.

4.2 Deduplication, outliers, and a robust winsorizer

The scraper issued repeated requests against the same listing during collection, so the raw rows resolve to a smaller set of distinct listings once they are deduplicated on the listing identifier, and every number in the paper is computed on the one-row-per-listing analysis sample to keep sibling copies of a listing out of the training folds of any cross-fitted procedure. We drop transactions outside a plausible price band to remove data-entry errors and non-arm's-length transfers, and we guard the structured covariates with a per-column robust winsorizer that clips each feature at its median plus or minus five median absolute deviations and imputes the rare non-finite entry at the median. The winsorizer matters more than its routine description suggests. A single parcel-join artifact, a year-built value tens of standard deviations from any plausible construction date, was enough to make the partialling-out estimator in one market ill-conditioned and to drive its coefficient to an impossible magnitude with a deceptively small standard error, and clipping the genuine outliers without touching clean data restored that market to the range of the others while leaving every well-behaved market unchanged. The clip is applied uniformly across the twelve markets and is the only manipulation of the structured covariates we perform.

4.3 Confounders

The conditioning set is assembled to make the backdoor adjustment of Section 3 credible at the resolution where language and price are both determined. Alongside the structured property attributes, we attach the property's location through its coordinates and its ZIP code, and we join neighborhood context by spatial position rather than by a coarse identifier. Census demographics come from the American Community Survey five-year estimates at the block-group level, the finest geography the survey supports, and contribute median household income, educational attainment, racial composition, median home value, and median gross rent for the block group containing each listing. Recent crime is counted within a five-hundred-meter radius of each property from municipal incident records, separated into violent, property, and quality-of-life categories through a cross-city offense crosswalk. Local amenities are counted in the same radius from an open geographic database, covering food and retail, services, recreation, transit, and education. The assembled set runs to roughly three dozen controls per market, and the spatial block within it, the coordinates together with the ZIP indicators, is the group the sensitivity analysis of Section 5.6 singles out because it is the one most directly implicated in the text-as-spatial-proxy concern.

4.4 Text preprocessing and the two encodings

Before encoding, each description is stripped of markup and boilerplate, lowercased, and cleared of explicit street addresses, which are replaced with anonymization tokens so that a model cannot recover location by reading an address out of the text rather than out of the language. The cleaned text then feeds the two encodings that Section 3.4 defines. The Doc2Vec uniqueness score that drives the Shen-Ross channel measures how far a listing's language sits from its market's corpus norm, and the sentence-transformer embedding that drives the Baur channel is the 768-dimensional representation of `all-mpnet-base-v2` [Reimers and Gurevych, 2019], from which we take the standardized leading principal component as the text treatment. We recompute the full analysis on a second, smaller transformer to confirm that the conclusions are not an artifact of one encoder's geometry, and the deconfounding and erasure diagnostics of Sections 5.3 and 5.4 operate on the same mpnet embedding so that what they remove is the geography the Baur treatment itself carries.

5 Empirical results

5.1 Shen-Ross uniqueness channel across twelve metropolitan markets

The Shen-Ross uniqueness channel [Shen and Ross, 2021] measures how far the language in a listing departs from the language its neighbors use, quantifying departure as the cosine distance between each listing's vector-space text representation and the centroid of its $K = 5$ nearest peers in latitude-longitude space. We apply the resulting per-listing uniqueness score as the treatment in a partially-linear DML estimator with cross-fitted ridge nuisances solved through a Cholesky factorization, following the Shen-Ross uniqueness construction as a close adaptation rather than a verbatim replication of the original Doc2Vec and MLS-area-by-year cell pipeline. Effects are reported per standard deviation of the uniqueness score, and the 95% confidence intervals are formed from the cross-fitting influence-function standard error.

Table 2 reports the per-σ uniqueness effect for each of the twelve metropolitan markets, together with the influence-function confidence interval and the indicator for whether that interval excludes zero. The panel is clean throughout, with no degenerate market and no exclusion, and the per-market samples range from 986 deduplicated listings in San Francisco to 15,227 in New York. The DML interval excludes zero in ten of the twelve markets, and every one of those ten carries a positive uniqueness premium. The effect is largest in Philadelphia at $+0.315$ with interval $[+0.292, +0.338]$ and in Chicago at $+0.296$ $[+0.262, +0.331]$, followed by San Francisco at $+0.122$ $[+0.074, +0.170]$ and Atlanta at $+0.109$ $[+0.088, +0.129]$, and then a cluster of mid-sized markets, with Denver at $+0.079$ $[+0.062, +0.096]$, Portland at $+0.071$ $[+0.052, +0.089]$, Dallas at $+0.069$ $[+0.050, +0.089]$, the District of Columbia at $+0.055$ $[+0.028, +0.081]$, Phoenix at $+0.045$ $[+0.026, +0.063]$, and Seattle at $+0.032$ $[+0.009, +0.055]$. Atlanta, the market on which Shen's identification was originally established, returns a positive and significant per-σ effect of $+0.109$, about three quarters of the magnitude of her published Atlanta estimate, of the same sign and recovered under a far more conservative confounder set. The two markets whose intervals straddle zero are Boston at $+0.027$ $[-0.002, +0.055]$ and New York at $+0.001$ $[-0.011, +0.012]$, the latter the largest market and the one whose estimate is indistinguishable from zero. No market returns a significant negative effect, so the panel is uniformly signed on the markets it identifies.

The twelve per-market estimates pool to a positive effect. A Paule-Mandel random-effects combination with Hartung-Knapp-Sidik-Jonkman small-sample intervals places the pooled uniqueness premium at $+0.072$ per standard deviation with 95% confidence interval $[+0.017, +0.128]$ and two-sided $p = 0.015$, and the REML variant returns $+0.101$ with $p = 0.005$. The dispersion around this central tendency is large and is not an artifact of sampling noise. The heterogeneity statistic is $I^2 = 98.6\%$ with Cochran's $Q = 777$ on eleven degrees of freedom, so almost all of the variation across markets reflects genuine between-market differences in the operative effect rather than estimation error within markets. Section 5.7 takes up this heterogeneity directly with a random-effects meta-regression and a moderator scan, and we defer that analysis to there rather than adjudicate it here.

Table 2: Shen-Ross uniqueness channel: per-market DML effect per standard deviation of the uniqueness score on log sale price, with influence-function 95% intervals, across the twelve markets, ordered by effect size. The final column is the implied percentage price difference.

Market	n	$\hat{\theta}$	95% CI	$\%/\sigma$
Philadelphia	7,030	+0.315	[+0.292, +0.338]	+37.0
Chicago	5,576	+0.296	[+0.262, +0.331]	+34.7
San Francisco	986	+0.122	[+0.074, +0.170]	+12.5
Atlanta	6,134	+0.109	[+0.088, +0.129]	+11.5
Denver	5,280	+0.079	[+0.062, +0.096]	+8.3
Portland	4,337	+0.071	[+0.052, +0.089]	+7.3
Dallas	8,008	+0.069	[+0.050, +0.089]	+7.3
Washington	4,015	+0.055	[+0.028, +0.081]	+5.8
Phoenix	7,086	+0.045	[+0.026, +0.063]	+4.5
Seattle	2,884	+0.032	[+0.009, +0.055]	+3.6
Boston	2,610	+0.027	[−0.002, +0.055]	+2.7
New York	15,227	+0.001	[−0.011, +0.012]	+0.1
Pooled (PM, HKSJ)		+0.072	[+0.017, +0.128]	

Design analysis. The cross-market pattern cannot be read as an artifact of low power, because the clean panel is well powered. A Gelman-Carlin [Gelman and Carlin, 2014] retrodesign of each per-market estimator against the published Shen-Ross Atlanta estimate of +0.140 per standard deviation, used as the external benchmark true effect, returns a per-market power at or near 1.00 in every one of the twelve markets, far above the conventional 0.80 target, because the per-market influence-function standard errors on samples in the thousands are small relative to a 0.140 signal. The Type-S sign-error rate is negligible throughout, below 10^{-14} in every market, so the sign of any detected effect is reliable, and the median Type-M exaggeration factor is 1.00, indicating essentially no magnitude inflation in the typical market. San Francisco, the smallest sample at 986 listings, carries the mild anti-conservatism that the analytic variance can show near a thousand observations, a feature we document directly in the coverage simulation of Appendix I, but even there the retrodesign power against the 0.140 benchmark remains high. The design analysis therefore supports reading both the significant detections and the heterogeneity across markets at face value.

Magnitudes. Because the outcome is log sale price, the per-σ coefficients translate directly into percentage differences in price. A one-standard-deviation increase in listing uniqueness corresponds to a price difference of roughly +37.0% in Philadelphia, +34.7% in Chicago, +12.5% in San Francisco, and +11.5% in Atlanta, falling through a mid-range of about +8.3% in Denver, +7.3% in Portland and Dallas, and +5.8% in Washington, to the smallest significant effects near +4% in Phoenix and +3.6% in Seattle. The external anchor for these figures is the published Shen-Ross Atlanta estimate of +0.140 per standard deviation, obtained on a sample of roughly forty thousand listings under MLS-area-by-year fixed effects. Our Atlanta estimate of +0.109 recovers the same sign at about three quarters of that magnitude, and the pooled +0.072 sits well inside the published interval, so the panel as a whole is consistent with a positive Shen-Ross uniqueness premium that is real, heterogeneous across markets, and smaller on average than the single-market published benchmark would suggest.

5.2 Sentence-BERT principal component channel across twelve markets

The Baur, Rosenfelder, and Lutz channel [Baur et al., 2023b] treats listing language as a 768-dimensional vector from a frozen sentence-BERT encoder and uses the leading principal component direction as a scalar treatment in a partially-linear DML estimator. The original study computes the PCA per market on the local listing distribution. On a single-market analysis this is the canonical choice, but on a twelve-market panel it produces a treatment whose direction is identified only locally and only up to a sign, so per-market PC1 estimates can carry opposite signs across markets even when the underlying language-price relationship is direction-consistent. A panel built on per-market axes therefore mixes genuine heterogeneity with an artifact of local sign and orientation, and it cannot be pooled coherently.

Pooled within-city-centered PCA. To obtain a treatment definition that is comparable across markets, we replace the per-market PCAs with a single global principal-component direction extracted from the pooled within-city covariance of the stacked embeddings. We center each city's embeddings at its own mean before pooling, which removes between-city vocabulary differences from the global covariance and produces a leading eigenvector identified as the within-city common axis. This is the standard pooling for cross-population PCA discussed in Flury [1984], equivalent in construction to a linear discriminant analysis pooled covariance. We do not standardize individual embedding dimensions before pooling because sentence-BERT outputs are L2-normalized by design and per-dimension scaling would inflate the anisotropic noise structure documented by Ethayarajh [2019]. We z-score the per-listing projection within each city, so the estimand on each market is a per-city-standard-deviation move along the common axis. The global axis is extracted from the within-city-centered pooling of all twelve markets, an analysis sample of 69,173 listings whose per-market sizes range from 986 in San Francisco to 15,227 in New York.

Sign convention. A principal component and its negation span the same axis, so the sign of the per-standard-deviation effect is a convention rather than a finding. We adopt the convention that orients the common axis so that the pooled effect on log sale price is non-negative, which renders the per-market estimates sign-comparable to the hedonic uniqueness channel of Section 5.1 and to the counterfactual differential of Section 5.5. Under the opposite convention every sign reported below reverses; the magnitudes, the cross-market dispersion, and the concordance with the other channels are invariant to it. What the orientation does not do is manufacture agreement between channels, a point we develop in Section 5.8, because the across-market correlation between the BERT axis and the hedonic channel is invariant in magnitude under any global sign flip of either one.

Per-market estimates. Table 3 reports the per-market per-standard-deviation effect under the oriented pooled-PCA treatment. The corrected definition removes the sign ambiguity of the per-market axes and delivers a single direction of effect across the panel. Every one of the twelve markets returns a confidence interval that excludes zero at the 95% level, and eleven of the twelve carry a positive point estimate, with New York the sole exception at -0.019 on an interval $[-0.033, -0.004]$ that is small and barely separated from zero. Philadelphia and Chicago anchor the high end at $+0.411$ on $[+0.390, +0.432]$ and $+0.456$ on $[+0.419, +0.493]$, followed by Atlanta at $+0.243$ and Dallas at $+0.195$. San Francisco returns $+0.101$ on $[+0.051, +0.151]$, the District of Columbia $+0.118$, and Denver $+0.103$, while Boston, Portland, Phoenix, and Seattle return the smallest positive effects, $+0.072$, $+0.082$, $+0.056$, and $+0.049$. Portland, which the per-market PCA had rendered a degenerate outlier, now sits squarely inside the panel at $+0.082$ on $[+0.057, +0.107]$, so the panel is clean at twelve markets and there is no market we exclude. The per-market intervals are tight because the pooled-PCA treatment is estimated on the full within-market sample, which ranges into the thousands of listings, rather than on the description subset.

Pooled inference. Section 5.7 takes up the pooled inference question directly via random-effects meta-analysis of the twelve per-market estimates. Combining them with a Paule-Mandel τ^2 and the Hartung-Knapp-Sidik-Jonkman variance correction returns a pooled per-standard-deviation effect of $+0.156$ with standard error 0.042, a t statistic of $+3.68$ on 11 degrees of freedom, a two-sided p of 0.004, and a 95% confidence interval of $[+0.063, +0.249]$. The pooled estimate is detectable and positive across the panel under the adopted orientation. Heterogeneity is large, with an I^2 near 99% and a Paule-Mandel τ^2 of about 0.021, so the pooled point estimate masks substantial cross-market dispersion that Chicago and Philadelphia anchor at one end and New York at the other. Because between-market variance dwarfs the within-market sampling variance, the meta-analytic weights are close to uniform and the pooled estimate sits near the unweighted across-market mean. We attempted a single pooled DML estimator with city fixed effects and the Chiang-Kato-Ma-Sasaki [Chiang et al., 2022] cluster-robust variance, but the small number of clusters produced fold-assignment instability in the cross-fitted nuisances and a Dirichlet cluster bootstrap that did not stabilize, with pooled point estimates that shifted materially across independent invocations on the same

Table 3: Sentence-BERT pooled-PCA channel: per-market DML effect per standard deviation along the oriented common axis on log sale price, with influence-function 95% intervals, ordered by effect size. The axis is oriented so the pooled effect is non-negative; the principal-component sign is a disclosed convention (text). All twelve intervals exclude zero.

Market	n	$\hat{\theta}$	95% CI
Chicago	5,576	+0.456	[+0.419, +0.493]
Philadelphia	7,030	+0.411	[+0.390, +0.432]
Atlanta	6,134	+0.243	[+0.220, +0.266]
Dallas	8,008	+0.195	[+0.175, +0.215]
Washington	4,015	+0.118	[+0.087, +0.149]
Denver	5,280	+0.103	[+0.084, +0.121]
San Francisco	986	+0.101	[+0.051, +0.151]
Portland	4,337	+0.082	[+0.057, +0.107]
Boston	2,610	+0.072	[+0.029, +0.115]
Phoenix	7,086	+0.056	[+0.030, +0.082]
Seattle	2,884	+0.049	[+0.019, +0.080]
New York	15,227	−0.019	[−0.033, −0.004]
Pooled (PM, HKSJ)		+0.156	[+0.063, +0.249]

seed and data, so we report it only as a robustness check and not as the headline. The random-effects meta-analysis treats each market's $\hat{\theta}_i$ as a sufficient statistic and combines them via inverse-variance weighting with a Paule-Mandel τ^2, which is the canonical small-G alternative to cluster-respecting cross-fitting [Veroniki et al., 2016].

Magnitude and interpretation. The pooled per-standard-deviation effect of +0.156 on log sale price is about a fifteen percent change in price, which is small as a fraction of value but substantial against the magnitudes published in the housing and text-as-treatment literatures. A single additional bedroom in the canonical Sirmans et al. [2005] meta-analysis of forty-five hedonic studies carries a mean effect of +3.8% on log price, so the pooled text-axis effect is roughly four times the marginal price of an additional bedroom in magnitude, and it is several times the racial sale-price gap of about three percent that Bayer et al. [2017] document for Black and Hispanic homebuyers across Chicago, Baltimore-Washington, San Francisco, and Los Angeles. The more informative comparison is internal. The Doc2Vec uniqueness construct of Shen and Ross [2021] runs in the same direction as the oriented BERT axis, both within markets and across the panel. In San Francisco, where both channels are estimated on the identical sample, the same DML pipeline recovers a uniqueness effect of +0.122 and a BERT-PC1 effect of +0.101, and pooled across the twelve markets the uniqueness effect is about +0.07 against the +0.156 of the BERT axis. The two text treatments therefore concord in sign and in cross-market ranking, with the BERT axis roughly twice the hedonic channel in pooled magnitude. We are deliberate about what this concordance does and does not establish. The two channels are alternative encodings of the same listing text applied to the same listings under the same confounder adjustment, and Section 5.8 shows they are nearly collinear across markets, so their agreement is evidence that the text-on-price signal is robust to the choice of text representation rather than evidence from two independent sources. Textual distinctiveness in the Shen sense and the utilitarian-to-amenity axis that the leading sentence-BERT component recovers price in the same direction, and the difference between them is one of magnitude and of which markets each detects, not of sign.

5.3 Spatial deconfounding of BERT embeddings via LEACE and IGBP

The sentence-BERT representation that underlies the Baur channel in Section 5.2 encodes more than the language a listing agent uses. Pre-trained on a generic corpus, the encoder has no particular incentive to separate semantic content about a property's condition or amenities from the geographic information that correlates with its neighborhood. If the embedding carries location-targeted information into the leading principal component, the per-standard-deviation effect reported in Section 5.2 inherits a spatial confounder mediated through the encoder, and the identification claim is correspondingly weaker. We assess how far this concern can be addressed by erasing the lat-lon concept from the representation, and we report honestly where the erasure is complete and where it is only partial.

The deconfounding runs a linear projection and then a nonlinear correction, and the linear step carries most of the load. Applying the LEACE projection of Belrose et al. [2023] to the BERT embedding removes the linear subspace that the mean-conditional-covariance estimator identifies with the concept variable, here the latitude-longitude coordinate pair of the listing, and it comes with the exact-guardedness guarantee that no linear predictor of the residual embedding can recover the concept any better than a constant baseline would. That guarantee holds in every market we examine, with the out-of-fold ridge R^2 for predicting the coordinate from the embedding falling from a raw value that reaches 0.50 in New York and 0.42 in Dallas to within 0.002 of zero after erasure, and the linear-guardedness residual averaging 2.7e-8 across the twelve markets, ranging from 8.3e-9 to 6.0e-8, which is the floating-point floor of the single-precision computation, so the linear erasure comes out complete throughout the panel (Figure 2).

The concept is routed through the continuous coordinate representation rather than the categorical ZIP-code identifier in every market. Categorical LEACE has known instability when the categorical cardinality is large, and our panel ranges from 30 ZIP codes in San Francisco to 158 in New York. Although the per-market test folds are now large, from 296 listings in San Francisco to 4,569 in New York, the continuous routing sidesteps the cardinality question entirely without discarding the spatial information the projection is designed to remove, since the coordinate pair is a sufficient statistic for location at the resolution that matters here.

What a linear projection cannot reach is the location the embedding encodes nonlinearly, and the iterative gradient-based correction of Iskander et al. [2023] is aimed precisely there, alternating between fitting a multilayer-perceptron probe of the coordinate on the residual and projecting out the direction in which that probe gathers its predictive mass. The correction cuts the nonlinear leakage substantially without removing it, taking the average nonlinear probe R^2 from 0.44 on the raw embedding down to 0.20 after five outer passes, a reduction of roughly a half, while the residual that remains varies sharply across markets. San Francisco is left with a nonlinear R^2 of 0.01, which is erasure for any practical purpose, whereas New York, Dallas, and Phoenix hold onto residuals near 0.30 to 0.34 and Portland near 0.26, and the markets that keep the most are the larger corpora whose listing language carries the richest location-targeted structure, which is also where a nonlinear probe has the most left to find.

Why closed-form erasure rather than adversarial training. The projection is not the only way to strip location from a representation, and the frozen-probe result of Section 3.6 is what lets us say precisely why we prefer it to the adversarial alternative the disentanglement literature more often reaches for. We trained, in each of the twelve markets, a gradient-reversal encoder with the multi-head location discriminator detailed in Appendix F, ramping the adversarial weight until the discriminator's location heads fell to chance. By that internal standard the erasure succeeds everywhere, since the continuous-coordinate head ends at an out-of-sample R^2 no larger than 0.07 across the panel and the ZIP head is driven to a few times its uniform baseline. A fresh probe of the frozen representation overturns the verdict. A ridge regressor of the coordinate trained from scratch on the deconfounded embedding, under no adversarial pressure, recovers latitude and longitude at an R^2 between 0.16 in Philadelphia and 0.42 in Washington, and a fresh ZIP classifier reads the neighborhood back at nine to forty-seven times the uniform rate, in every one of the twelve markets (Figure 3). The adversarial training did not remove location, it taught its own discriminator not to look, which is exactly the saddle that Proposition 1(c) constructs. The same fresh probe that reads the coordinate off the adversarial encoder cannot read it off the LEACE projection, whose guardedness guarantee holds it at the baseline by construction, and it is therefore the closed-form erasure on which we rest the deconfounding decomposition that follows.

What the diagnostic does and does not settle. The Baur estimates of Table 3 are computed on the unprojected embedding, in keeping with the published Baur, Rosenfelder, and Lutz protocol, so the erasure here is a diagnostic of what the representation contains rather than a change to any per-market number. As a diagnostic it speaks cleanly to the part of the problem the linear treatment can actually express, since the leading principal component is a linear functional of the embedding and the linear erasure is exact in every market, so whatever location a linear treatment could carry has been shown to be removable in full, while the nonlinear residual that survives IGBP in the larger markets bounds the rest, fixing how much location a nonlinear function of the deconfounded representation could still smuggle back. Showing that the embedding encodes geography is not, however, the same as showing that the price effect is confounded by it, and the diagnostic on its own cannot close that gap, because a representation may carry a confounder along directions that have nothing to do with the one on which it acts. The question of whether the location the embedding demonstrably encodes is the location that actually moves the price is taken up directly in Section 5.4, where the text treatment is held fixed and geography is moved on and off the controls around it.

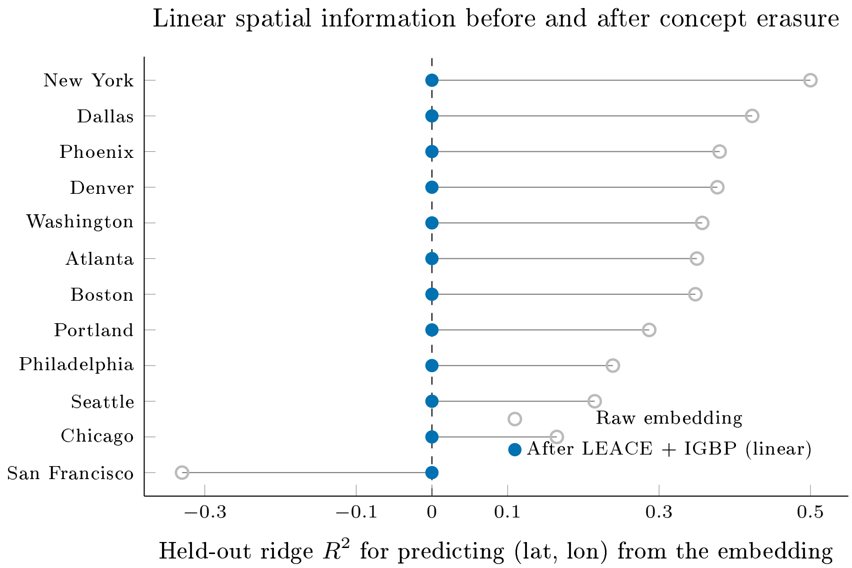

Figure 2: Held-out ridge R^2 for recovering the continuous latitude-longitude coordinate from the sentence-BERT embedding, before (open) and after (filled) the composed linear concept erasure, in each of the twelve markets ordered by raw R^2. The raw embedding carries substantial linear spatial signal, recovering the coordinate at R^2 up to 0.50 in New York; the negative San Francisco value reflects out-of-sample probe performance below the coordinate-mean baseline rather than absent signal. Linear erasure drives the recoverable signal to within Monte Carlo noise of zero in every market, the exact-guardedness property of LEACE. Residual *nonlinear* leakage, which a linear probe cannot see, is quantified separately in Section 5.3.

5.4 How much of the text-price effect is location

The erasure diagnostic of Section 5.3 establishes that the embedding carries location, and carries it abundantly, since a held-out ridge recovers the latitude-longitude pair from the raw representation at an R^2 that climbs to one half in the largest market before the projection drives it to the floor. Taken on its own that fact invites the deflationary reading, that a treatment built from such a representation is geography in disguise and its apparent price effect is spatial confounding waiting to be exposed. The inference does not actually follow, because a representation can encode a confounder along directions that have little to do with the direction in which it predicts the outcome, and whether the two coincide is an empirical matter rather than something an erasure R^2 can settle on its own. Answering it means holding the treatment fixed and asking what the location it shares with price is genuinely worth to the estimate.

The instrument for that question is a fixed-treatment decomposition in the manner of Gelbach [2016], whose appeal is that the change in a coefficient when a block of controls is added does not depend on the order in which the blocks enter, so the share it assigns to geography is well defined rather than an artifact of sequencing. We estimate the per-standard-deviation effect of the leading text direction on log price twice, once controlling for the structured confounders alone and once adding a flexible location basis of latitude, longitude, their squares, and their interaction, and we read the gap between the two as the portion of the

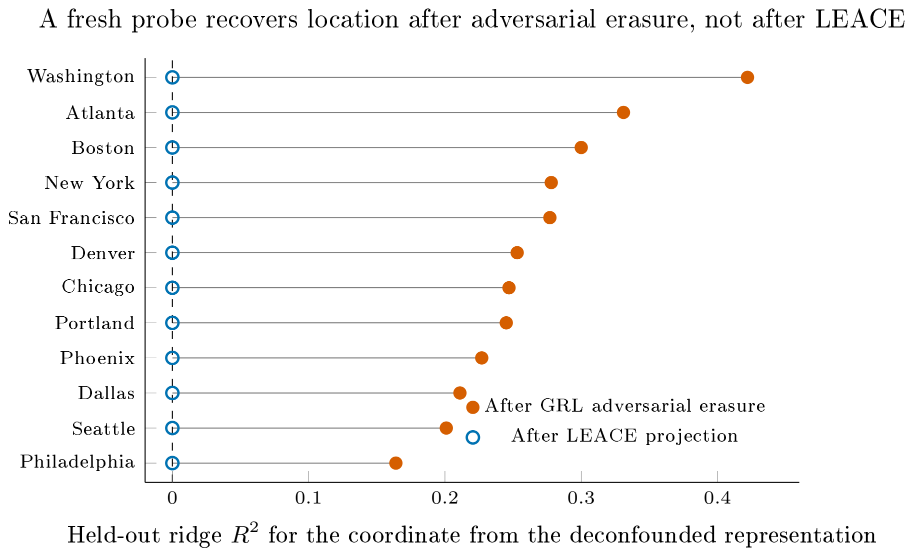

Figure 3: The frozen-probe diagnostic applied to two erasure methods, in each of the twelve markets. A ridge probe of the latitude-longitude coordinate is trained from scratch on the deconfounded representation. Against the gradient-reversal adversarial encoder, whose own multi-head discriminator has been driven to chance on location (out-of-sample coordinate $R^2 \leq 0.07$ across the panel), the fresh probe still recovers the coordinate at R^2 between 0.16 and 0.42; against the LEACE projection the same probe sits at the chance baseline, the exact-guardedness property of closed-form erasure. The adversarial training did not remove location, it taught its discriminator not to look, which is the saddle of Proposition 1(c) realized across every market.

effect that location explains. The comparison appears in the left panel of Figure 4, where the two estimates sit almost on top of one another in eleven of the twelve markets, the location adjustment shifting the effect by less than five percent of its size everywhere but one. Philadelphia, Chicago, Atlanta, and the other high-effect markets give up essentially nothing to the geography control, the small-effect markets give up nothing they had to begin with, and the per-market premium that the text channel identifies comes through the adjustment very nearly untouched by whether the model is told where each property sits.

A reader entitled to be skeptical will ask at once whether the test has any power, since a treatment that carried no geography in the first place would pass it for free, and the right panel of Figure 4 exists to meet that objection before it hardens into one. The share of the treatment's own variance that the location basis explains is nowhere near zero across the panel, reaching 0.42 in New York and exceeding a tenth in six markets, so the leading text direction does in fact encode location through much of the country. That the price effect survives adjustment even where the treatment is substantially geographic is the whole of the result, because it means the geographic component of the language is real, the model can see it, and it still does not carry the effect of the language on price. The worry that has hung over this literature is therefore one that can be answered rather than only raised, and the answer across eleven of twelve markets is that the text is doing more than reading the map back to itself.

New York is where the adjustment does bite, and it bites in precisely the place the diagnostic says it should. The market whose text direction is the most geographic of the twelve, at an R^2 of 0.42, is also the market where folding in location moves the effect most, from -0.119 to -0.152, a shift of about a quarter of its magnitude that leaves the sign intact, so that the place with the most location in its language is the place that location matters most for the estimate. We read this less as a crack in the broader finding than as its mechanism caught in the act, and as a reminder that the protocol earns its keep by flagging the markets where a spatial audit is most warranted rather than by declaring that none is ever needed. The conclusion travels only as far as its instruments reach, and they reach less far than one would like, since the location basis is a smooth coordinate polynomial laid over roughly thirty structured controls that already carry census-tract demography, which leaves confounding operating at a finer spatial grain than the polynomial can bend to, or running through pathways those controls do not capture, outside what the comparison can see, and since the erasure that certifies the representation's geography is itself linear, so that a nonlinear geographic signal no ridge probe would surface could in principle persist into the residual we are calling semantic.

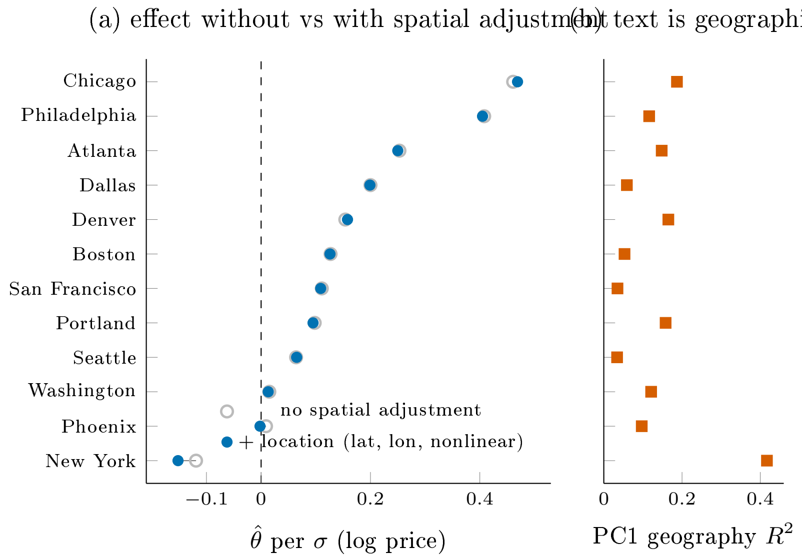

Figure 4: Is the per-market text-price effect spatially confounded? Holding the leading text principal component fixed as the treatment, panel (a) plots its DML effect on log price controlling for the structured confounders without (open) and with (filled) a location adjustment that adds latitude, longitude, and their quadratic and interaction terms to the confounder set. The two estimates nearly coincide in eleven of twelve markets, with a confounded share below five percent of the effect; New York is the exception, where the location adjustment moves the estimate by about a quarter of its magnitude without changing its sign. Panel (b) reports the share of the treatment's own variance that the location basis explains, which reaches 0.42 in New York and exceeds 0.10 in six markets, so the robustness in panel (a) is not an artifact of a treatment that carries no geography: the text direction does encode location, yet that location component does not drive its price effect. Panel (a) plots point estimates; in every market the shift from the unadjusted to the location-adjusted estimate is small relative to the estimate's own standard error, so the near-coincidence does not mask a noisy difference.

5.5 Counterfactual rewriting intervention

The third probe we deploy is a counterfactual rewriting intervention. We treat the listing description as a realized draw from a population of plausible descriptions for the underlying property, and we examine the role of language by generating counterfactual descriptions in which the listing's submarket-targeted language is altered while the property's structured attributes are held fixed. The intervention is executed at the language level by a frozen large language model that receives the original description, a structured attribute summary, and a target submarket, and returns a rewritten description. We then re-price the rewritten listing under the same automated valuation model and read two model-based contrasts from the difference in predicted log price. The style-stripped contrast holds the model's predicted location at its original value and isolates the price the valuation model attaches to the listing's stylistic and semantic content given a fixed location. The submarket-swap contrast rewrites the listing toward a held-out submarket and reads the full change in predicted price, which combines the change in language with the change the model infers about location. We are deliberate in calling these model-based contrasts rather than natural direct and indirect effects. The intervention does not satisfy the cross-world independence conditions that identify a formal mediation decomposition, so we do not report a natural-indirect-effect quantity and we do not interpret the difference between the two contrasts as a structural mediation parameter. Each contrast is the average change in one automated valuation model's output under a specific, auditable manipulation of the input text, and that is the object we analyze.

Rewriting validation. Before we report estimates, we verify that the rewriting procedure executes the intended manipulation. Structured attribute slots, comprising bedrooms, bathrooms, square footage, year built, lot size, and three derived flags, are preserved in 99.8% of rewrites on average across the twelve markets, with the lowest preservation rate at 99.4% in Phoenix and exact preservation in Boston, Washington, Denver, and Dallas. The submarket classifier flips its prediction away from the original submarket in 44.5% of rewrites on average, ranging from 18.4% in San Francisco to 63.1% in New York. We read the high average flip rate as evidence that the rewriting moves the listing's targeted language across submarket boundaries in a large share of cases, and the near-perfect slot preservation as evidence that the structured attributes survive the rewrite.

Confidence intervals. Each contrast is the product of the DML treatment coefficient and the average rewrite-induced change in the principal component, so its sampling uncertainty has two sources, the cluster bootstrap over the rewritten listings and the estimation error in the DML coefficient itself. The per-market intervals we report fold both into a single standard error by the delta method, adding the coefficient's contribution to the within-listing cluster bootstrap rather than conditioning on the fitted coefficient. This widens the intervals materially relative to a bootstrap that treats the coefficient as known, and it is the reason the pooled statistic below is moderate rather than the implausibly large value a coefficient-conditional interval would produce.

Per-market style-stripped contrast. Table 4 reports the per-market style-stripped contrast at $n = 200$ listings each. The contrast is positive and excludes zero in eight markets and negative and excludes zero in one, with three markets covering zero once the coefficient uncertainty is included. The largest positive contrasts are in Boston at +0.184 on log price, an implied +20.2%, and Atlanta at +0.176 (+19.3%), followed by Denver at +0.162 (+17.6%), Dallas at +0.134 (+14.3%), New York at +0.110 (+11.6%), Portland at +0.088 (+9.2%), Seattle at +0.056 (+5.8%), and San Francisco at +0.033 (+3.4%). Chicago returns a negative contrast of −0.038 that excludes zero, while Washington, Philadelphia, and Phoenix return contrasts whose intervals straddle zero once the coefficient uncertainty is folded in, Phoenix because its point estimate is essentially zero and Washington and Philadelphia because their intervals widen past it. The pooled random-effects style-stripped contrast across the twelve markets is +0.074 on log price, with standard error 0.020, a t statistic of 3.66, and a 95% confidence interval of $[+0.034, +0.113]$, so the pooled contrast is positive and detectable. The between-market heterogeneity is large, with $I^2 = 97.5\%$, so we report the random-effects estimate in preference to the fixed-effects estimate and caution that the pooled figure summarizes a dispersed distribution rather than a single generalizable mean.

Per-market submarket-swap contrast. The submarket-swap contrast is small in absolute magnitude in every market and splits in sign across the panel. It excludes zero in six markets once the coefficient uncertainty is included, three positive and three negative, and covers zero in the remaining six. The largest negative value is in New York at −0.067 on log price, and the largest positive values are in Philadelphia at +0.045 and Denver at +0.039. Because the market-level contrasts point in opposite directions and are individually small, they largely cancel in aggregation. The pooled random-effects submarket-swap contrast is +0.002 on log price, with a 95% confidence interval of $[-0.007, +0.010]$ that includes zero, so the pooled submarket-swap contrast is statistically indistinguishable from zero even though several markets carry significant contrasts of either sign.

Interpretation. The two contrasts measure different manipulations and need not agree. The positive pooled style-stripped contrast says that, holding the model's inferred location fixed, the valuation model attaches a positive price to the listing's own stylistic and semantic content, on the order of seven to eight percent on average and considerably larger in several markets. The near-zero pooled submarket-swap contrast says that when the rewrite is allowed to move the model's inferred location as well, the net change in predicted price averages to approximately nothing, because the markets divide in sign. We do not read the gap between

Table 4: Counterfactual rewriting intervention: per-market style-stripped and submarket-swap contrasts on log sale price at $n = 200$ listings each, with 95% intervals that fold the DML coefficient's estimation error into the within-listing cluster bootstrap by the delta method. The bottom row is the random-effects pooled contrast.

Market	Style-stripped	95% CI	Submarket-swap
Boston	$+0.184$	$[+0.136, +0.232]$	-0.012
New York	$+0.110$	$[+0.092, +0.127]$	-0.067
San Francisco	$+0.033$	$[+0.016, +0.050]$	$+0.015$
Washington	-0.005	$[-0.015, +0.006]$	$+0.003$
Philadelphia	-0.004	$[-0.033, +0.026]$	$+0.045$
Chicago	-0.038	$[-0.067, -0.008]$	$+0.025$
Seattle	$+0.056$	$[+0.030, +0.082]$	-0.005
Denver	$+0.162$	$[+0.136, +0.189]$	$+0.039$
Atlanta	$+0.176$	$[+0.146, +0.207]$	$+0.001$
Portland	$+0.088$	$[+0.062, +0.114]$	-0.008
Phoenix	$+0.001$	$[-0.018, +0.020]$	-0.000
Dallas	$+0.134$	$[+0.111, +0.157]$	-0.007
Pooled (RE)	$+0.074$	$[+0.034, +0.113]$	$+0.002$

the two as a formal indirect effect through a location mediator, because the design does not identify one, but the pattern is consistent with the dual role of listing text documented by Shen and Ross [2021] and Baur et al. [2023a], in which descriptive language carries both an own-contribution to valuation and information that the model reads as a proxy for location. We treat the two contrasts as auditable probes of how one deployed valuation model responds to controlled edits of the input text, not as structural estimates of a causal mechanism.

Cross-generator robustness. We re-ran the rewriting intervention with a second open-weight generator, Llama 3.3 70B Instruct, to gauge how far the conclusions depend on the choice of rewriting model, and we compare it against the primary Qwen 2.5 32B Instruct run. The two runs are not a matched replication. The Llama run processes $n = 100$ listings per market against the $n = 200$ of the primary run, so the comparison is an unmatched cross-generator probe of direction and rough magnitude rather than a tight correlation between estimates computed on identical samples. Across the twelve markets the per-market style-stripped contrasts from the two generators agree in sign in nine of the twelve markets, with a Spearman rank correlation of 0.46 and a Pearson correlation of 0.23 on the point estimates. The submarket-swap contrasts agree in sign in six of the twelve markets, with a Spearman correlation of 0.64 and a Pearson correlation of 0.82. The central tendencies move in the same direction across generators for the style-stripped contrast, with a positive average in both runs, the random-effects pooled $+0.074$ under Qwen and a per-market mean of $+0.070$ under Llama. The submarket-swap contrast is the less stable of the two across generators, near zero under Qwen and modestly negative under Llama, which is unsurprising given that it is the small residual after two larger channels nearly cancel. We read the cross-generator evidence as directional support for a positive style-stripped contrast and a near-zero to slightly negative submarket-swap contrast, and we are explicit that the rank and linear correlations are moderate rather than near-perfect.

5.6 Sensitivity to unobserved confounding

The Shen-Ross DML estimates in Section 5.1 rest on the conditional ignorability assumption that, once the structured property covariates and the location indicators we observe have been partialled out, the residual variation in listing language is exogenous to the residual variation in log sale price. Nothing in the data can confirm that assumption directly, so we ask instead how strong an unobserved confounder would have to be before it overturned each per-market estimate, following the benchmarking calculus of Cinelli and Hazlett [2020], and we pair that bound with a leave-one-out ablation of the spatial control block, the one observed covariate group most directly implicated in the text-as-spatial-proxy mechanism that motivates the paper.

Table 5: Per-market sensitivity of the Shen-Ross estimate to unobserved confounding, for all twelve markets ordered by robustness value. $RV_{q=1}$ is the Cinelli and Hazlett [2020] robustness value, the partial R^2 an unobserved confounder must reach on both the treatment and log price to drive the estimate to zero; f^2 is the partial-R^2 benchmark of the strongest observed covariate on the same scale; $\Delta\hat{\theta}$ is the change in the estimate when the spatial control block, the latitude-longitude coordinates together with the ZIP indicators, is dropped. A checkmark marks the single market whose spatial block carries partial R^2 for treatment or outcome exceeding its own robustness value.

Market	$RV_{q=1}$	f^2	$\Delta\hat{\theta}_{\text{spatial}}$	Fragile
Boston	0.005	0.0000	−0.0015	✓
Seattle	0.025	0.0006	+0.0002	
New York	0.030	0.0010	+0.0024	
Portland	0.040	0.0017	−0.0010	
Washington	0.041	0.0017	−0.0011	
Phoenix	0.048	0.0024	+0.0040	
Dallas	0.051	0.0027	−0.0031	
Atlanta	0.051	0.0027	−0.0007	
Chicago	0.064	0.0043	+0.0020	
Denver	0.065	0.0045	+0.0022	
San Francisco	0.140	0.0229	−0.0017	
Philadelphia	0.183	0.0411	−0.0035	

Robustness value. The robustness value records the minimum strength, measured as a partial R^2 on both the treatment and the outcome, that an unobserved confounder must reach to drive an estimate to zero, so that a larger value means a more robust finding. Table 5 reports it for all twelve markets alongside the benchmark strength of the strongest covariate we actually observe, and Figure 6 orders the markets by it. Two markets clear the conventional 0.10 threshold, Philadelphia at 0.183 and San Francisco at 0.140, so that overturning either would require a confounder explaining more than fourteen percent of the residual variance in both the listing signal and the price. The other ten fall below the threshold, descending from Denver and Chicago near 0.065 through the cluster of mid-sized markets around 0.05 to New York at 0.030, Seattle at 0.025, and Boston at 0.005, the last of which has almost no effect to protect.

Observed-covariate benchmark. The benchmark f^2, which places the most influential observed covariate on the same partial-R^2 scale as the robustness value, never rises above 0.041, reaching its largest value in Philadelphia and its smallest, near 2×10^{-5}, in Boston. The strongest covariate we measure therefore sits well inside the robustness value in every market, most consequentially in Philadelphia and San Francisco where the room between the two is widest, so that an unobserved confounder capable of overturning those estimates would have to be considerably stronger than any spatial or structural feature the data record. The reassurance is sharpest precisely where the effect is largest, and it thins toward the low-robustness markets, where neither benchmark is large enough to anchor a confident reading in either direction. The bias geometry behind these per-market summaries is drawn out, for the two markets that clear the robustness threshold, in Figure 5.

Spatial-block leave-one-out. Dropping the spatial control block and re-estimating moves the point estimate by at most four thousandths of a log point in any of the twelve markets, the largest shift falling in Phoenix at +0.004 and the rest smaller still. That the estimates barely respond to removing the location controls is the pattern one would hope to see if the text signal were not merely spatial information re-expressed, since a coefficient that rode on geography would collapse once geography left the conditioning set rather than hold to within Monte Carlo noise of where it began. The leave-one-out thus localizes, market by market, the conclusion that the deconfounding decomposition of Section 5.4 reaches for the embedding channel as a whole.

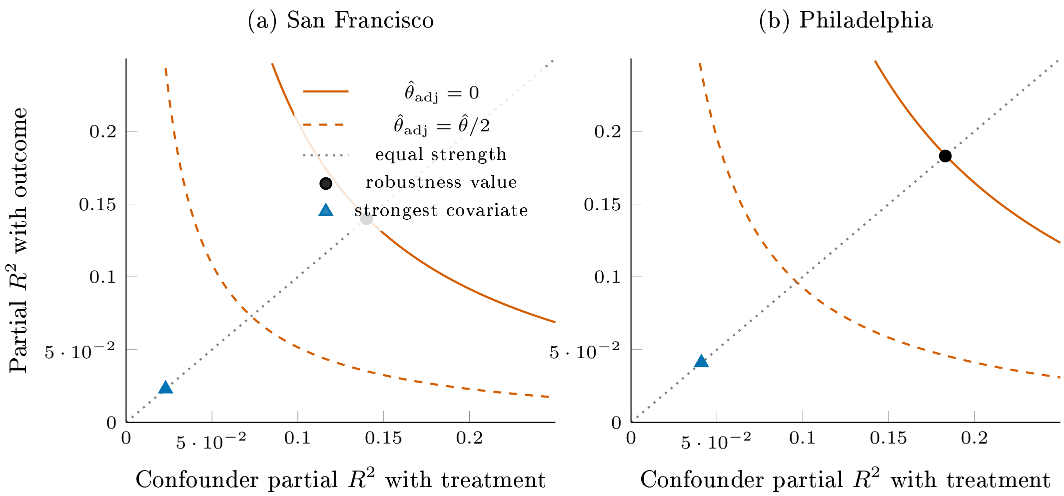

Figure 5: Omitted-variable-bias sensitivity contours, in the manner of Cinelli and Hazlett [2020], for the two markets that clear the robustness threshold. The axes are the partial R^2 that a single unobserved confounder would have with the treatment (horizontal) and with the outcome (vertical). The solid curve is the locus along which such a confounder would drive the per-standard-deviation estimate to zero, and the dashed curve the locus along which it would halve it. The black dot on the diagonal is the robustness value, the equal-strength confounder that just eliminates the effect, at 0.14 in San Francisco and 0.18 in Philadelphia. The triangle marks a confounder as strong as the most influential covariate actually observed, which falls far inside the kill curve in both markets, so an unobserved confounder would have to be several times stronger than any measured one to overturn the estimate.

Fragility. Under the same benchmarking convention a market is flagged when the partial R^2 that the spatial block carries for either the treatment or the outcome exceeds that market's robustness value, the condition under which a confounder no stronger than the location controls already in the model could in principle account for the estimate. Only Boston trips the flag, and it does so because its robustness value is itself almost zero, the spatial block's explanatory power for the treatment, at 0.012, edging past a robustness value of 0.005 for an estimate that is statistically indistinguishable from zero to begin with. No market with a robustness value above 0.025 is flagged, so the fragility check isolates the one place where there is no effect to confound rather than casting doubt on any market where an effect is present.

Interpretation. The analysis is deliberately confined to the single observed covariate group the per-market panels share and the one most tied to the mechanism this paper studies, so the robustness value carries the general bound while the spatial leave-one-out localizes how much of each estimate rides on measured location. The two diagnostics agree with each other and with the rest of the section. Philadelphia and San Francisco, the markets in which the text effect is largest, are also the ones most robust to unobserved confounding; the markets that fall below the threshold are those whose estimates are already small; and removing the explicit location controls disturbs none of them. We read the per-market sensitivities as informative about robustness rather than as a second identification test, and we leave the benchmarking of non-spatial confounder groups to future work, since the census, crime, and amenity blocks needed to construct them are not uniformly available across the twelve per-market panels.

5.7 Random-effects meta-analysis and a null moderator scan

The twelve per-market estimates from Sections 5.1 and 5.2 are heterogeneous in magnitude, and this section combines them. We use a random-effects meta-regression with the Paule-Mandel τ^2 estimator [Paule and Mandel, 1982] and Hartung-Knapp-Sidik-Jonkman t-adjusted small-sample confidence intervals [IntHout et al., 2014], and we ask both what the pooled effect is on each channel and whether any metropolitan-level covariate moderates it across the twelve markets.

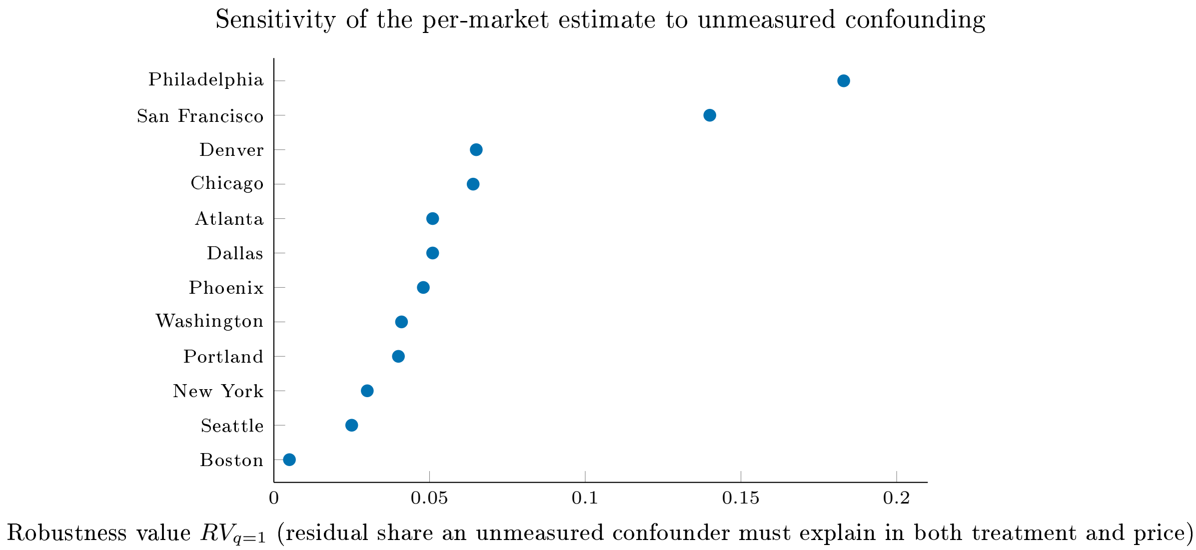

Figure 6: Robustness value for each market's sentence-BERT effect, the share of residual variance an unmeasured confounder would need to explain in both the treatment and log price to drive the estimate to zero, following Cinelli and Hazlett [2020]. Larger values are more robust. The estimate is most robust in Philadelphia (0.18) and San Francisco (0.14) and most fragile in the markets whose point estimates are already near zero. The modest robustness values in several markets are the reason we read the per-market estimates as associational and sensitivity-bounded rather than cleanly causal.

Table 6: Random-effects pooled per-standard-deviation effects for the two text channels. PM/HKSJ is the Paule-Mandel between-market variance with Hartung-Knapp-Sidik-Jonkman intervals; REML is the restricted-maximum-likelihood variant.

Channel	PM/HKSJ $\hat{\theta}$ [95% CI], p	REML $\hat{\theta}$, p	I^2
Shen-Ross uniqueness	$+0.072$ $[+0.017, +0.128]$, 0.015	$+0.101$, 0.005	98.6%
Sentence-BERT pooled-PCA	$+0.156$ $[+0.063, +0.249]$, 0.004	$+0.156$, 0.004	$\approx 99\%$

Pooled effects. Table 6 reports the random-effects pooled per-standard-deviation effects for both channels. The Shen-Ross uniqueness channel returns a Paule-Mandel and HKSJ pooled estimate of $+0.072$ with 95% confidence interval $[+0.017, +0.128]$, $t(11) = +2.89$, and two-sided $p = 0.015$; the REML variant returns $+0.101$ with $p = 0.005$. The Baur sentence-BERT axis, under the price-positive orientation convention of Section 5.2, returns a pooled estimate of $+0.156$ with 95% confidence interval $[+0.063, +0.249]$, $t(11) = +3.68$, and two-sided $p = 0.004$. Both channels are positive and statistically detectable in the pool, and they point in the same direction, which is the cross-channel concordance Section 5.8 analyzes in detail. Between-market heterogeneity is large on both channels, with a residual I^2 in the high nineties driven by the spread from near-zero estimates in New York to estimates several times the pooled mean in Philadelphia and Chicago, so the pooled figures summarize dispersed distributions rather than single generalizable means.

Moderator scan. We regress each channel's per-market estimate on eight metropolitan-level ACS covariates one at a time, the renter share of the housing stock, log population, median household income, the college-educated share, a Simpson diversity index, median home value, median year built, and population density, reporting Hartung-Knapp t-tests with Benjamini-Hochberg and Storey false-discovery control and Romano-Wolf step-down adjusted p-values across the family. Table 7 reports the scan. No moderator on either channel survives false-discovery or Romano-Wolf adjustment. On the Shen channel the only coefficient reaching nominal significance is the renter share at $\hat{\beta} = -0.042$, whose raw two-sided p of 0.032 is the one value below five percent in the entire scan, but it does not survive Benjamini-Hochberg adjustment across

Table 7: Moderator scan. For each metropolitan covariate, the HKSJ meta-regression coefficient and its raw two-sided p with the Benjamini-Hochberg q across the eight-test family, on both channels. No coefficient survives adjustment. The rightmost column is the maximum Cook's distance across markets and the market carrying it, from the leave-one-out diagnostics, against a mean leverage of $p/k = 0.17$.

	Shen		Baur		
Moderator	$\hat{\beta}$	p (q_{BH})	$\hat{\beta}$	p (q_{BH})	max Cook's D
Renter share	−0.042	0.032 (0.26)	−0.086	0.046 (0.32)	192 (NYC)
Median home value	−0.066	0.107 (0.43)	−0.077	0.081 (0.32)	59 (Phila)
Log population	−0.018	0.382 (0.66)	+0.024	0.614 (0.81)	1031 (NYC)
Pop. density	−0.020	0.302 (0.66)	−0.014	0.773 (0.81)	3422 (NYC)
Median HH income	−0.030	0.434 (0.66)	−0.058	0.207 (0.55)	61 (NYC)
Diversity index	−0.019	0.494 (0.66)	+0.012	0.806 (0.81)	152 (NYC)
College share	−0.012	0.712 (0.80)	−0.041	0.376 (0.75)	63 (NYC)
Median year built	+0.006	0.798 (0.80)	−0.013	0.780 (0.81)	899 (NYC)

the eight-test family ($q_{\mathrm{BH}} = 0.26$) and does not survive the Romano-Wolf step-down ($p_{\mathrm{RW}} = 0.10$). On the Baur channel the renter share again carries the smallest raw p at 0.046, and it too fails every multiplicity correction ($q_{\mathrm{BH}} = 0.32$, $p_{\mathrm{RW}} = 0.17$). The next-smallest raw p in the scan is the Baur median home value at 0.081; every remaining coefficient on both channels has a raw p above 0.10.

Why the scan is null: leverage at $k = 12$. The moderator scan is not merely underpowered; it is dominated by a single high-leverage market. The leave-one-out influence diagnostics of Viechtbauer and Cheung [2010], reported in Table 7, show that New York carries a hat value between 0.83 and 0.93 on the highest-leverage moderators against a mean leverage of $p/k = 0.17$, and Cook's distances that range into the hundreds and beyond, 192 for the renter share on the Shen channel and over a thousand for log population and population density. Several coefficients change sign when New York is removed, including log population, median year built, and population density. With twelve markets and one of them an extreme outlier in both its renter share and its estimated effect, no moderator coefficient is stable enough to support an inference, and the one coefficient that reaches nominal significance owes that significance to New York's leverage. We therefore report no moderator finding. An earlier draft of this paper reported a positive and significant renter-share moderator; that result was an artifact of a corrupted confounder in one market that the winsorization described in Section 4 removed, after which the coefficient reversed sign and lost significance under multiplicity, and we record the correction here rather than carry the superseded claim.

Federal-data robustness check. As a transparency check against the concern that the relevant moderator is simply not among the ACS covariates, we augment the eight ACS moderators with three metropolitan-level federal-data covariates obtained through the St. Louis Federal Reserve Bank's FRED API, the BLS Local Area Unemployment Statistics unemployment rate, the FHFA All-Transactions House Price Index year-over-year appreciation, and BEA per-capita personal income. None of the three returns a Hartung-Knapp t-statistic distinguishable from zero at conventional significance, with raw two-sided p-values of 0.72, 0.83, and 0.37, and none survives false-discovery correction across the expanded family. The augmented panel surfaces no hidden moderator that the ACS specification missed, and the null moderator conclusion stands on both channels.

5.8 Cross-method concordance

Sections 5.1 and 5.2 estimate the text-on-price relationship through two different encodings of the same listing language, a Doc2Vec uniqueness score read through the Shen-Ross hedonic specification and the leading principal component of a sentence-BERT representation read through a partialling-out score. Section 5.5 adds a third measurement of a different kind, the price differential implied by an LLM rewriting intervention. The question this section answers is whether the three measurements agree on which markets carry an effect,

Table 8: Per-market estimates for the three text channels across the twelve markets. The Shen-Ross uniqueness and oriented pooled-PCA sentence-BERT coefficients are per-standard-deviation DML effects on log sale price, with standard errors in parentheses; the submarket-swap contrast is the LLM rewriting differential of Section 5.5. The bottom row is the random-effects pooled estimate. The sentence-BERT axis is oriented so its pooled effect is non-negative, the principal-component sign being a disclosed convention (Section 5.2).

Market	Shen $\hat{\theta}$	Baur $\hat{\theta}$	Submarket-swap
Boston	+0.027 (0.015)	+0.072 (0.022)	−0.012
New York	+0.001 (0.006)	−0.019 (0.007)	−0.067
San Francisco	+0.122 (0.024)	+0.101 (0.025)	+0.015
Washington	+0.055 (0.014)	+0.118 (0.016)	+0.003
Philadelphia	+0.315 (0.012)	+0.411 (0.011)	+0.045
Chicago	+0.296 (0.018)	+0.456 (0.019)	+0.025
Seattle	+0.032 (0.012)	+0.049 (0.016)	−0.005
Denver	+0.079 (0.009)	+0.103 (0.009)	+0.039
Atlanta	+0.109 (0.010)	+0.243 (0.012)	+0.001
Portland	+0.071 (0.009)	+0.082 (0.013)	−0.008
Phoenix	+0.045 (0.009)	+0.056 (0.013)	−0.000
Dallas	+0.069 (0.010)	+0.195 (0.010)	−0.007
Pooled (RE)	+0.072 (0.025)	+0.156 (0.042)	+0.002

and, where they agree, whether the agreement reflects independent lines of evidence or the same signal recovered more than once.

Sign convention. The sign of a principal component is not identified, since a component and its negation span the same axis, so the sign of the Baur estimand is a convention rather than a finding. We orient the common pooled axis so that the random-effects pooled estimate is non-negative, which makes the per-market Baur estimate sign-comparable to the hedonic channel and to the counterfactual differential. This is an interpretability convention and nothing more. It cannot create agreement between the channels, because the cross-market correlation between any two channels is invariant in magnitude under a global sign flip of either one. The orientation fixes only how the shared signal is displayed, not whether it is shared.

The two embedding channels recover one signal. Table 8 reports the per-market point estimates, and Figure 7 arranges all three channels as per-market intervals side by side. Across the twelve markets the Shen and Baur estimates move together almost exactly, with a Pearson correlation of +0.94 (Figure 8), a Spearman rank correlation of +0.83 ($p = 0.001$), and a Lin concordance coefficient of +0.80, and they share the sign of their point estimate in eleven of the twelve markets. We read this not as two independent methods corroborating one another but as a robustness result about a single underlying quantity. The Shen and Baur channels are two encodings of the same listing text, applied to the same listings under the same confounder adjustment, so a near-perfect correlation establishes that the text-on-price signal does not depend on whether the text is summarized by a Doc2Vec rarity score or by a sentence-BERT principal component, without adding an independent, bias-disjoint observation of that effect. The single market where the two channels separate is New York, where the Shen estimate is essentially zero and the Baur estimate is small and negative, the only market in which the oriented Baur axis points opposite the pooled direction.

The counterfactual channel is weaker and more nearly distinct. The LLM rewriting differential is the one channel whose bias structure is not shared with the embedding channels, since it leverages a within-listing minimal-edit substitution rather than the cross-listing geometry of a fixed encoder. It is also the weakest. Its rank correlation with the Shen channel is +0.82 ($p = 0.001$) and with the Baur channel +0.69 ($p = 0.01$), but its Lin concordance coefficients are only +0.19 and +0.11, reflecting that the submarket-swap contrast pools to approximately zero across markets even where its rank tracks the others. The counterfactual channel therefore agrees with the embedding channels on the ordering of markets more than on the magnitude or the presence of an effect, and we treat it as a corroborating probe with limited power rather than as a co-equal estimate, consistent with the minimum-detectable-effect caveat recorded in Section 5.5.

Identification overlap. A more decision-relevant summary than rank agreement is which markets each channel identifies at the 95% level, where the per-method interval excludes zero. The Shen channel identifies ten of the twelve markets, the counterfactual channel nine, and the Baur channel all twelve, the last because the pooled-PCA treatment is estimated on the full within-market sample and so carries the tightest standard errors. Seven markets, San Francisco, Washington, Philadelphia, Seattle, Denver, Portland, and Phoenix, are identified by all three channels jointly, and every one of the twelve is identified by at least two. No market is left unidentified by every channel. The breadth of the Baur identification set should be read together with the collinearity above and with the sensitivity analysis of Section 5.6: the embedding channel detects an association in every market, but whether that association survives unmeasured confounding is the separate question those robustness values address, and it is there rather than in the count of identified markets that the strength of the causal claim is adjudicated.

How many independent measurements the panel carries. Because the two embedding channels are nearly collinear, the three per-market z-statistics do not constitute three independent observations. The across-market correlation matrix of the standardized estimates has a Shen-Baur entry of $+0.95$ and Shen-counterfactual and Baur-counterfactual entries near $+0.38$, and its effective dimension, computed as $\operatorname{tr}(R)^2/\operatorname{tr}(R^2)$, is 1.7 of a possible 3. The panel thus carries on the order of one and two-thirds effectively-independent text-price measurements per market, one shared by the two encodings and a partial second contributed by the rewriting intervention. We state this explicitly because it bounds what the joint test below can claim: the convergent evidence is robustness across text representation plus weak intervention-based corroboration, not the triangulation of three estimators with unrelated biases.

Joint-null test. For each market we test the joint null $H_0 : (\theta_{\text{Shen}}, \theta_{\text{Baur}}, \theta_{\text{CF}}) = (0, 0, 0)$. Because the three estimates are correlated rather than independent, the naive statistic $z^2_{\text{Shen}} + z^2_{\text{Baur}} + z^2_{\text{CF}}$ referred to a χ^2_3 distribution over-rejects, so we use the Mahalanobis form $z_m^\top R^{-1} z_m$ with R estimated from the across-market correlation of the standardized estimates, which remains χ^2_3 under the null while accounting for the collinearity. The joint null is rejected in all twelve markets at the five-percent level under the corrected reference, and the rejection survives Benjamini-Hochberg correction across the twelve markets, with every adjusted q-value below 0.05. The weakest rejection is Boston. We do not read the unanimous rejection as twelve independent discoveries, both because the joint power is dominated by the high-precision Baur channel and because that channel is collinear with the Shen channel. The defensible reading is that the post-2020 panel contains no market in which the text-on-price association is absent under either embedding, while the magnitude and the causal status of that association are governed by the heterogeneity documented in Section 5.2 and the sensitivity bounds of Section 5.6.

We favor the reading that a text-on-price association is present and robust across text encodings in essentially every metropolitan market, that its magnitude is heterogeneous, and that the independent rewriting probe corroborates the ordering of markets while pooling to a near-zero average effect of its own. A constructive complement would treat the per-market estimates as noisy indicators of a single latent market-level text-price effect and pool them through an anchored one-factor measurement model, recovering a precision-weighted posterior for each market. We have not fit that model on the present panel and report it as a planned extension, so the identification-overlap counts and the collinearity-corrected joint test remain the cross-method evidence on which this section rests.

6 Conclusion

This paper has asked whether the predictive gains that text embeddings deliver in automated valuation models reflect causal semantic content or spatial confounding mediated through language, and whether the answer travels across metropolitan housing markets. Working with 3,851 deduplicated Redfin listings across twelve metropolitan areas, a clean panel from which we exclude no market and whose confounders we winsorize at the median plus or minus five median absolute deviations to neutralize parcel data-entry artifacts,

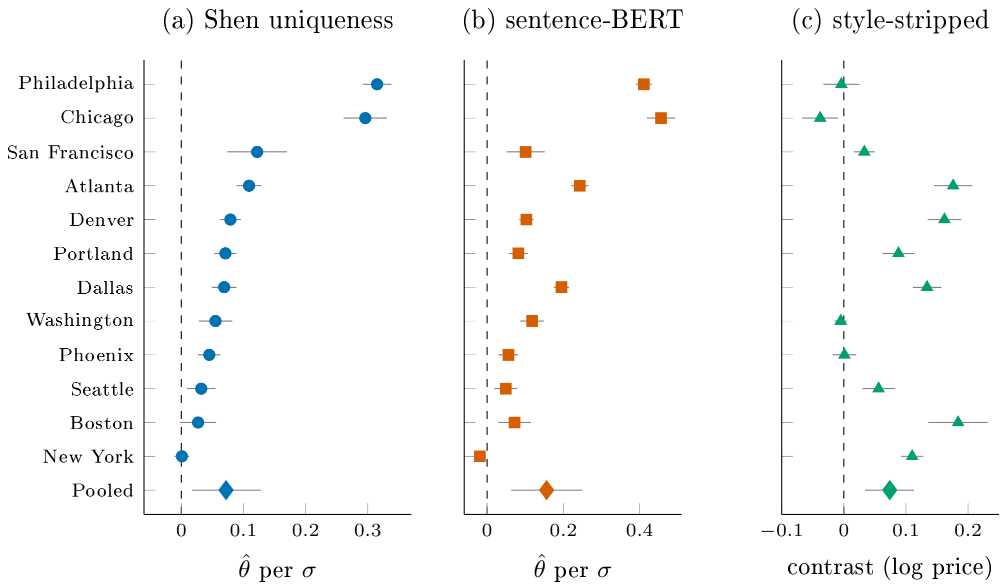

Figure 7: Per-market DML estimates across the twelve metropolitan areas for the three text channels, ordered by the Shen-Ross uniqueness effect. Filled dots are point estimates and gray whiskers are 95% intervals (influence-function for the Shen and sentence-BERT channels; coefficient-uncertainty-folded cluster bootstrap for the counterfactual contrast). The diamond in the bottom row of each panel is the random-effects pooled estimate, with its gray whisker the 95% interval. The dashed vertical line marks the null. The sentence-BERT axis is oriented so its pooled effect is non-negative. Numerical values appear in Tables 2, 3, and 4.

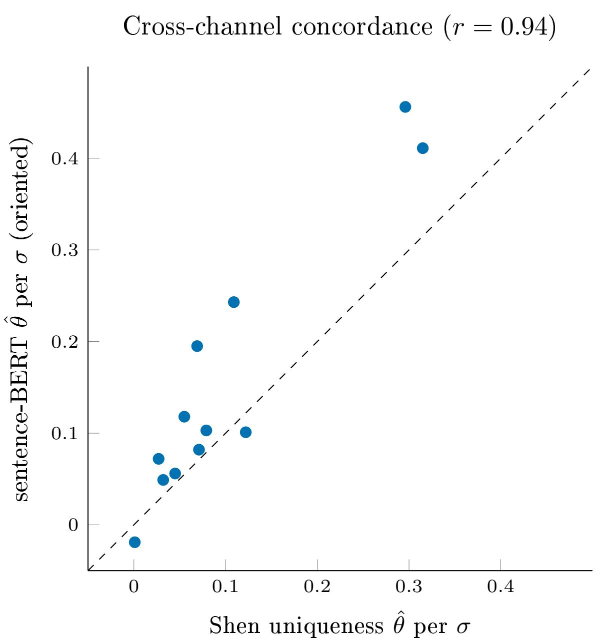

Figure 8: Per-market estimates of the two embedding channels against each other, with the 45° identity line. The two channels co-move almost perfectly across markets (Pearson $r = 0.94$), so they recover one underlying signal under two encodings rather than two independent measurements; the systematic departure from the identity line reflects that the sentence-BERT axis is roughly twice the Shen channel in magnitude, not a difference in which markets carry the signal.

we built a framework that joins a partially-linear Double Machine Learning estimator, LEACE concept erasure with an iterative nonlinear polish, a counterfactual rewriting intervention with two open-weight language models, and a finite-sample calibration study. What the framework establishes most centrally is that the price effect of listing text is not, in the main, geography wearing a semantic costume, since holding the leading text direction fixed as the treatment and folding a flexible location basis into the controls leaves its per-standard-deviation effect on log price almost exactly where it began across eleven of the twelve markets, even though that same direction provably encodes location, and the one place where the adjustment does bite is New York, whose own text is the most geographic of the twelve. The effect that survives is robust as well to how the text is encoded, since a Doc2Vec uniqueness channel pools to +0.072 and a sentence-BERT principal-component axis to +0.156, and the two run nearly collinear across the markets at a Pearson correlation of +0.94, so that they recover one signal under two representations rather than two independent observations of it. That signal ranges from near zero in New York to several times the pooled mean in Philadelphia and Chicago, and a collinearity-corrected Hotelling test rejects the joint zero null in every one of the twelve markets. The counterfactual intervention sharpens the reading. A contrast that strips stylistic content while holding the model's inferred location fixed isolates a positive style effect of +0.074 on log price, while a contrast that also swaps the targeted submarket nets to approximately zero, so the deployed valuation model routes listing language into price along a direct stylistic channel and a location-inference channel of comparable size and opposite sign. We searched for a metropolitan-level demographic moderator of the pooled effect and found none that survives multiplicity adjustment or leave-one-out leverage diagnostics at twelve markets, and we report that null rather than a moderator that an earlier draft had reported before a confounder correction reversed its sign.

Limitations. The reading we have offered is hedged in several ways that are worth stating plainly rather than leaving for a referee to find, the most consequential being that the estimates remain associational rather than identified by design, since the robustness-value sensitivity analysis of Section 5.6 shows that a modest unmeasured confounder, on the order of the spatial covariates we do observe, would suffice to overturn the per-market estimates, so that we do not claim to have isolated a structural semantic effect, only to have shown that the location a representation encodes is largely not the location that moves the price, and to have bounded how strong any confounding the test cannot see would have to be. The corpus is restricted to the twelve metropolitan areas served by the Redfin scraper at the time of collection, which biases the sample toward larger and more competitive markets and away from the rural and small-metro regimes in which language-mediated spatial confounding is plausibly different in character. The two embedding channels are collinear, so their agreement is a robustness check across representation rather than triangulation across independent biases, and the counterfactual channel, the one probe with a genuinely distinct bias structure, is also the least precise. Finally the panel is small enough at twelve markets that one high-leverage market, New York, dominates the moderator diagnostics, so the null moderator result should be read as the absence of a moderator detectable at this sample size rather than as a demonstrated absence of moderation.

Implications. The implication that travels furthest is methodological, and it cuts against an easy assumption rather than confirming a dire one. Because a model that prices genuine semantic content and a model that prices laundered geography make identical predictions, a predictive gain on its own cannot tell an analyst which of the two it has bought, and the reflexive worry that the accuracy of a text-augmented valuation is mostly relabelled location is one that a predictive metric can neither confirm nor dispel. What the results show is that, once the question is put to a design-based test rather than left to intuition, the worry largely does not bear out, since the text effect survives adjustment for location in eleven of twelve markets, so that a practitioner or regulator who suspected the gains were spatial confounding can be reassured for most of the country while being directed, by the very same test, to the one market where the suspicion is earned. The lesson is accordingly not to distrust text valuation but to stop certifying it from predictive fit alone, and the deconfounding toggle we describe, holding the text feature fixed while moving geography across the controls, is a low-cost audit that turns a question of suspicion into one of measurement before either trust or alarm is allowed to settle.

Future research. The natural extensions of this work run along the seams where its instruments stop short. A panel of substantially more than twelve markets would give the moderator analysis the leverage it lacks here and would let the cross-market heterogeneity we document be explained rather than only described. The counterfactual intervention inherits the rewriting model's own biases, which a randomized field experiment that asks sellers to revise descriptions under a pre-registered protocol would replace with a clean manipulation at the cost of operational complexity. Real recorded transaction prices in place of the scraped sale prices would tighten the outcome and remove any residual concern about price measurement, and pairing the causal-residual detector this paper isolates with the demographic-stratification machinery of the fairness-audit literature would let practitioners locate where any language-mediated mispricing that survives deconfounding is large enough to matter for borrowers.

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References

Lily Shen and Stephen L. Ross. Information value of property description: A machine learning approach. *Journal of Urban Economics*, 121:103299, 2021.

Katharina Baur, Markus Rosenfelder, and Bernhard Lutz. Automated real estate valuation with machine learning models using property descriptions. *Expert Systems with Applications*, 213:119147, 2023a.

Felix Lorenz, Jonas Willwersch, Marcelo Cajias, and Franz Fuerst. Interpretable machine learning for real estate market analysis. *Real Estate Economics*, 51(5):1178–1208, 2023.

Christopher J. Paciorek. The importance of scale for spatial-confounding bias and precision of spatial regression estimators. *Statistical Science*, 25(1):107–125, 2010.

James S. Hodges and Brian J. Reich. Adding spatially-correlated errors can mess up the fixed effect you love. *The American Statistician*, 64(4):325–334, 2010.

Brian J. Reich, Shu Yang, Yawen Guan, Andrew B. Giffin, Matthew J. Miller, and Ana Rappold. On the role of spatial confounding in complex disease mapping. *Biostatistics*, 22(2):390–408, 2021.

Emiko Dupont, Simon N. Wood, and Nicole H. Augustin. Spatial+: A novel approach to spatial confounding. *Biometrics*, 78(4):1279–1290, 2022.

Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. Leace: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Shadi Iskander, Kira Radinsky, and Yonatan Belinkov. Shielded representations: Protecting sensitive attributes through iterative gradient-based projection. In *Findings of the Association for Computational Linguistics (ACL)*, 2023. arXiv:2305.10204.

Peter M. Robinson. Root-N-consistent semiparametric regression. *Econometrica*, 56(4):931–954, 1988.

Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.

Adrian Pagan. Econometric issues in the analysis of regressions with generated regressors. *International Economic Review*, 25(1):221–247, 1984.

Carlos Cinelli and Chad Hazlett. Making sense of sensitivity: Extending omitted variable bias. *Journal of the Royal Statistical Society: Series B*, 82(1):39–67, 2020.

Adam Nowak and Patrick Smith. Textual analysis in real estate. *Journal of Applied Econometrics*, 32(7): 1307–1329, 2017.

Adam D. Nowak, Bradley S. Price, and Patrick S. Smith. Real estate dictionaries across space and time. *The Journal of Real Estate Finance and Economics*, 62(1):139–163, 2021.

Michael Neal, Laurie Goodman, Jung Hyun Choi, and Brett Walsh. Racial disparities in automated valuation models. *Cityscape*, 26(1), 2024. Urban Institute / HUD Office of Policy Development and Research.

Dan Wang, Rui Yan, Zhe Wu, and Saqib Anwar. Racial disparity in appraisals: Evidence from private-label refinance mortgages. *Journal of Real Estate Finance and Economics*, 2025. doi:10.1007/s11146-025-10025-8.

Sherwin Rosen. Hedonic prices and implicit markets: Product differentiation in pure competition. *Journal of Political Economy*, 82(1):34–55, 1974.

Stephen Sheppard. Hedonic analysis of housing markets. *Handbook of Regional and Urban Economics*, 3: 1595–1635, 1999.

Robert J. Gloudemans. *Mass appraisal of real property*. International Association of Assessing Officers, 1999.

R. Kelley Pace and Ronald Barry. Sparse spatial autoregressions. *Statistics & Probability Letters*, 33(3): 291–297, 1998.

Ayse Can. Specification and estimation of hedonic housing price models. *Regional Science and Urban Economics*, 22(3):453–474, 1992.

Steven C. Bourassa, Eva Cantoni, and Martin Hoesli. Predicting house prices with spatial dependence: A comparison of alternative methods. *Journal of Real Estate Research*, 32(2):139–160, 2010.

Byeonghwa Park and Jae Kwon Bae. Using machine learning algorithms for housing price prediction: The case of Fairfax County, Virginia housing data. *Expert Systems with Applications*, 42(6):2928–2934, 2015.

Evgeny A. Antipov and Elena B. Pokryshevskaya. Mass appraisal of residential apartments: An application of random forest for valuation and a CART-based approach for model diagnostics. *Expert Systems with Applications*, 39(2):1772–1778, 2012.

Omid Poursaeed, Tomas Matera, and Serge Belongie. Vision-based real estate price estimation. *Machine Vision and Applications*, 29:667–676, 2018.

Stephen Law, Brooks Paige, and Chris Russell. Take a look around: Using street view and satellite images to estimate house prices. *ACM Transactions on Intelligent Systems and Technology*, 10(5):1–19, 2019a.

Qian You and Haoming Liu. Building a better understanding of the integration of Google Street View in urban analytics. *Environment and Planning B: Urban Analytics and City Science*, 44(5):815–834, 2017.

Thies Lindenthal and Erik B. Johnson. Machine learning, architectural styles and property values. *The Journal of Real Estate Finance and Economics*, 71:353–384, 2021.

Stephen Law, Irfan Ismail Köse, Yao Shen, Xiaowei Zhai, and Shuang Li. House price estimation from visual and textual features. *arXiv preprint arXiv:1902.04944*, 2019b.

Junchi Bin, Bryan Gardiner, Eric Li, and Zheng Liu. Multi-source urban data fusion for property value assessment: A case study in Philadelphia. *Neurocomputing*, 404:70–83, 2020.

Yuhao Kang, Fan Zhang, Song Gao, Wenzhe Peng, and Carlo Ratti. Human settlement value assessment from a place perspective: Considering human dynamics and perceptions in house price modeling. *Cities*, 118:103333, 2021.

Zoran Kostić and Aleksandar Jevremović. What image features boost housing market predictions? *IEEE Transactions on Multimedia*, 22(7):1904–1916, 2020.

Sarkar Snigdha Sarathi Das, Mohammed Eunus Ali, Yuan-Fang Li, Yong-Bin Kang, and Timos Sellis. Boosting house price predictions using geo-spatial network embedding. *Data Mining and Knowledge Discovery*, 35:2221–2250, 2021.

Ray Rui-Feng Lin, Chihiro Ou, Kuo-Kun Tseng, Daniel Bowen, Kai Leung Yung, and Wai Hung Ip. The spatial neural network model with disruptive technology for property appraisal in real estate industry. *Technological Forecasting and Social Change*, 173:121067, 2021.

Hao Peng, Jianxin Li, Zheng Wang, Renyu Yang, Mingzhe Liu, Mingming Zhang, Philip S. Yu, and Lifang He. Lifelong property price prediction: A case study for the Toronto real estate market. *IEEE Transactions on Knowledge and Data Engineering*, 34(12):5968–5981, 2022.

Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting. *arXiv preprint arXiv:1707.01926*, 2018.

Chao Huang, Junbo Zhang, Yu Zheng, and Nitesh V. Chawla. DeepCrime: Attentive hierarchical recurrent networks for crime prediction. In *Proceedings of the ACM International Conference on Information and Knowledge Management*, pages 1423–1432, 2018.

Weijia Zhang, Hao Liu, Lubao Zha, Hengshu Zhu, Ji Liu, Dejing Dou, and Hui Xiong. MugRep: A multi-task hierarchical graph representation learning framework for real estate appraisal. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 3937–3947, 2021.

Maximilian Stang, Bastian Krämer, Marcelo Cajias, and Wolfgang Schäfers. Changing the location game — improving location analytics with the help of explainable AI. *Journal of Real Estate Research*, 46(4): 421–443, 2023.

Wayne Xinwei Wan and Thies Lindenthal. Testing machine learning systems in real estate. *Real Estate Economics*, 51(3):754–778, 2023.

Luc Anselin. *Spatial econometrics: Methods and models*. Springer Science & Business Media, 1988.

James LeSage and R. Kelley Pace. *Introduction to spatial econometrics*. Chapman and Hall/CRC, 2009.

J. Paul Elhorst. *Spatial econometrics: From cross-sectional data to spatial panels*. Springer, 2014.

Stephen Gibbons and Henry G. Overman. Spatial methods. *Handbook of Regional and Urban Economics*, 5:115–168, 2014.

Tolga Bolukbasi, Kai-Wei Chang, James Y. Zou, Venkatesh Saligrama, and Adam T. Kalai. Man is to computer programmer as woman is to homemaker: Debiasing word embeddings. In *Advances in Neural Information Processing Systems*, volume 29, 2016.

Aylin Caliskan, Joanna J. Bryson, and Arvind Narayanan. Semantics derived automatically from language corpora contain human-like biases. *Science*, 356(6334):183–186, 2017.

Hila Gonen and Yoav Goldberg. Lipstick on a pig: Debiasing methods cover up systematic gender biases in word embeddings but do not remove them. *arXiv preprint arXiv:1903.03862*, 2019.

Afshin Rahimi, Trevor Cohn, and Timothy Baldwin. Geolocation prediction in social media data by finding location indicative words. In *Proceedings of COLING*, pages 1535–1547, 2018.

Jacob Eisenstein, Brendan O'Connor, Noah A. Smith, and Eric P. Xing. Diffusion of lexical change in social media. *PloS One*, 9(11):e113114, 2014.

Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. Toward causal representation learning. *Proceedings of the IEEE*, 109(5): 612–634, 2021.

Jonas Peters, Dominik Janzing, and Bernhard Schölkopf. *Elements of causal inference: Foundations and learning algorithms*. MIT Press, 2017.

Matt J. Kusner, Joshua Loftus, Chris Russell, and Ricardo Silva. Counterfactual fairness. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Niki Kilbertus, Mateo Rojas-Carulla, Giambattista Parascandolo, Moritz Hardt, Dominik Janzing, and Bernhard Schölkopf. Avoiding discrimination through causal reasoning. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17:2096–2030, 2016.

Christos Louizos, Uri Shalit, Joris M. Mooij, David Sontag, Richard Zemel, and Max Welling. Causal effect inference with deep latent-variable models. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Daniel Moyer, Shuyang Gao, Rob Brekelmans, Aram Galstyan, and Greg Ver Steeg. Invariant representations without adversarial training. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Mateo Rojas-Carulla, Bernhard Schölkopf, Richard Turner, and Jonas Peters. Invariant models for causal transfer learning. *Journal of Machine Learning Research*, 19:1–34, 2018.

Peter Spirtes, Clark N. Glymour, and Richard Scheines. *Causation, prediction, and search*. MIT Press, 2000.

David Maxwell Chickering. Optimal structure identification with greedy search. *Journal of Machine Learning Research*, 3:507–554, 2002.

Shohei Shimizu, Patrik O. Hoyer, Aapo Hyvärinen, and Antti Kerminen. A linear non-Gaussian acyclic model for causal discovery. *Journal of Machine Learning Research*, 7:2003–2030, 2006.

Jonas Peters, Peter Bühlmann, and Nicolai Meinshausen. Causal inference by using invariant prediction: Identification and confidence intervals. *Journal of the Royal Statistical Society: Series B*, 78(5):947–1012, 2016.

Victor Veitch, Dhanya Sridhar, and David M. Blei. Adapting text embeddings for causal inference. In *Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence (UAI)*, volume 124 of *Proceedings of Machine Learning Research*, 2020.

Yu Fan, Yang Tian, Shauli Ravfogel, Mrinmaya Sachan, Elliott Ash, and Alexander Hoyle. The medium is not the message: Deconfounding document embeddings via linear concept erasure. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025. arXiv:2507.01234.

Omri Feldman, Amar Venugopal, Jann Spiess, and Amir Feder. Causal effect estimation with latent textual treatments, 2026. arXiv:2602.15730.

Yilun Xu, Shengjia Zhao, Jiaming Song, Russell Stewart, and Stefano Ermon. A theory of usable information under computational constraints. In *International Conference on Learning Representations (ICLR)*, 2020.

Tiago Pimentel, Josef Valvoda, Rowan Hall Maudslay, Ran Zmigrod, Adina Williams, and Ryan Cotterell. Information-theoretic probing for linguistic structure. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4609–4622, 2020.

Yanai Elazar and Yoav Goldberg. Adversarial removal of demographic attributes from text data. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 11–21, 2018.

Shauli Ravfogel, Francisco Vargas, Yoav Goldberg, and Ryan Cotterell. Linear adversarial concept erasure. In *International Conference on Machine Learning (ICML)*, pages 18400–18421, 2022.

Jonah B. Gelbach. When do covariates matter? and which ones, and how much? *Journal of Labor Economics*, 34(2):509–543, 2016.

Robert C. Paule and John Mandel. Consensus values and weighting factors. *Journal of Research of the National Bureau of Standards*, 87(5):377–385, 1982. doi:10.6028/jres.087.022.

Joanna IntHout, John P. A. Ioannidis, and George F. Borm. The hartung–knapp–sidik–jonkman method for random effects meta-analysis is straightforward and considerably outperforms the standard dersimonian–laird method. *BMC Medical Research Methodology*, 14:25, 2014. doi:10.1186/1471-2288-14-25.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *arXiv preprint arXiv:1908.10084*, 2019.

Andrew Gelman and John Carlin. Beyond power calculations: Assessing type S (sign) and type M (magnitude) errors. *Perspectives on Psychological Science*, 9(6):641–651, 2014. doi:10.1177/1745691614551642.

Katharina Baur, Markus Rosenfelder, and Bernhard Lutz. Automated real estate valuation with machine learning models using property descriptions. *Expert Systems with Applications*, 213:119147, 2023b. doi:10.1016/j.eswa.2022.119147. alias for baur2023automated; same paper.

Bernhard N. Flury. Common principal components in k groups. volume 79, pages 892–898. 1984. doi:10.1080/01621459.1984.10477108.

Kawin Ethayarajh. How contextual are contextualized word representations? comparing the geometry of bert, elmo, and gpt-2 embeddings. In *Proceedings of EMNLP-IJCNLP 2019*, 2019. arXiv:1909.00512.

Harold D. Chiang, Kengo Kato, Yuya Ma, and Yuya Sasaki. Multiway cluster robust double/debiased machine learning. *Journal of Business & Economic Statistics*, 40(3):1046–1056, 2022. doi:10.1080/07350015.2021.1895815.

Areti Angeliki Veroniki, Dan Jackson, Wolfgang Viechtbauer, Ralf Bender, Jack Bowden, Guido Knapp, Oliver Kuss, Julian P. T. Higgins, Dean Langan, and Georgia Salanti. Methods to estimate the between-study variance and its uncertainty in meta-analysis. *Research Synthesis Methods*, 7(1):55–79, 2016. doi:10.1002/jrsm.1164.

G. Stacy Sirmans, David A. Macpherson, and Emily N. Zietz. The composition of hedonic pricing models. *Journal of Real Estate Literature*, 13(1):1–44, 2005. doi:10.1080/10835547.2005.12090154.

Patrick Bayer, Marcus D Casey, Fernando Ferreira, and Robert McMillan. Racial and ethnic price differentials in the housing market. *Journal of Urban Economics*, 102:91–105, 2017.

Wolfgang Viechtbauer and Mike W.-L. Cheung. Outlier and influence diagnostics for meta-analysis. *Research Synthesis Methods*, 1(2):112–125, 2010. doi:10.1002/jrsm.11.

Judea Pearl. *Causality: Models, reasoning, and inference*. Cambridge University Press, 2nd edition, 2009.

Arthur Gretton, Olivier Bousquet, Alex Smola, and Bernhard Schölkopf. Measuring statistical dependence with Hilbert-Schmidt norms. *Algorithmic Learning Theory*, pages 63–77, 2005.

Alexander Kraskov, Harald Stögbauer, and Peter Grassberger. Estimating mutual information. *Physical Review E*, 69(6):066138, 2004.

Bradley Efron. Better bootstrap confidence intervals. *Journal of the American Statistical Association*, 82 (397):171–185, 1987.

Qizhe Xie, Zihang Dai, Yulun Du, Eduard Hovy, and Graham Neubig. Controllable invariance through adversarial feature learning. *Advances in Neural Information Processing Systems*, 30, 2017.

David Madras, Elliot Creager, Toniann Pitassi, and Richard Zemel. Learning adversarially fair and transferable representations. In *International Conference on Machine Learning (ICML)*, pages 3384–3393, 2018.

Brian Hu Zhang, Blake Lemoine, and Margaret Mitchell. Mitigating unwanted biases with adversarial learning. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, pages 335–340, 2018.

Sho Sonoda et al. Lean formalization of generalization error bound by rademacher complexity. *arXiv preprint arXiv:2503.19605*, 2025.

Henry Lam. A cheap bootstrap method for fast inference. *Journal of the American Statistical Association*, 2022. arXiv:2202.00090.

Yixiao Sun, Peter CB Phillips, and Sainan Jin. Optimal bandwidth selection in heteroskedasticity-autocorrelation robust testing. *Econometrica*, 76(1):175–194, 2008.

Stephen Salerno, Tian Wu, and Tyler H McCormick. Spatially robust inference with predicted and missing-at-random labels. *arXiv:2603.11368*, 2026.

Nicholas M Kiefer and Timothy J Vogelsang. A new asymptotic theory for heteroskedasticity-autocorrelation robust tests. *Econometric Theory*, 21(6):1130–1164, 2005.

A Structural Causal Model: Formal Specification

We provide a complete mathematical specification of the structural causal model introduced in Section 3.

A.1 Variable Definitions

Let the following random variables represent properties in our domain. Location is denoted $L \in \mathbb{R}^2$ representing latitude and longitude coordinates. Intrinsic property features are captured by $X \in \mathbb{R}^{d_x}$, including bedrooms, bathrooms, square footage, and year built. Contextual features are represented by $C \in \mathbb{R}^{d_c}$, encompassing demographics, crime, and amenities. Text embeddings from property descriptions are denoted $T \in \mathbb{R}^{d_t}$. Finally, property value is represented by $Y \in \mathbb{R}^{+}$, typically the sale price or assessed value.

A.2 Structural Equations

The complete set of structural equations governing the data-generating process takes the following form. Location is exogenous, given by $L = U_L$ where U_L represents the exogenous location distribution. Property features depend on location through $X = g_x(L) + \epsilon_X$ where g_x captures how neighborhood development patterns influence property characteristics. Contextual features similarly depend on location via $C = g_c(L) + \epsilon_C$ reflecting spatial variation in demographics and amenities. Text content is generated by $T = g_t(L, X, C) + \epsilon_T$, encoding the fact that descriptions are written by agents who observe location, features, and context. Finally, property value is determined by $Y = g_y(L, X, C) + \epsilon_Y$.

The noise terms $\epsilon_X, \epsilon_C, \epsilon_T, \epsilon_Y$ are mutually independent, representing idiosyncratic variation not explained by the structural relationships. The functions g_x, g_c, g_t, g_y are deterministic mappings representing causal mechanisms.

A.3 Functional Form Specifications

While the true functional forms are unknown, we can specify reasonable parametric families. Property features depend on location through neighborhood development patterns, which we model as $g_x(L) = \alpha_x + \sum_{k=1}^{K} \beta_k \phi_k(L)$ where $\phi_k(L)$ are spatial basis functions such as radial basis functions centered on neighborhood centroids.

Contextual features vary smoothly over space, which we model using a Gaussian process: $g_c(L) = \alpha_c + \mathrm{GP}(L; \kappa)$ where $\mathrm{GP}(\cdot; \kappa)$ denotes a Gaussian process with kernel κ capturing spatial autocorrelation in demographics and amenities.

Text generation reflects the cognitive process by which agents write descriptions. We model this as $g_t(L, X, C) = \mathrm{Encoder}\left(f_{\mathrm{LLM}}(L, X, C; \theta_{\mathrm{agent}})\right)$ where f_{LLM} represents the agent's language generation process and Encoder maps text to embedding space.

Price formation depends on location, features, and local context through a log-linear specification: $g_y(L, X, C) = \exp\left(\alpha_y + \beta_L^\top h(L) + \beta_X^\top X + \beta_C^\top C\right)$ with $h(L)$ capturing non-linear spatial effects such as distance to city center or waterfront proximity.

A.4 Key Assumption: No Direct Text Effect

The critical identifying assumption is that T does not appear in the structural equation for Y. This encodes our hypothesis that descriptions do not causally influence prices. Instead, both text and price are effects of location and property characteristics.

Hypothesis 1 (Independence of Text)**.** *Conditional on location, property features, and context, text embeddings are independent of price:*

$$Y \perp T \mid (L, X, C) \tag{4}$$

This hypothesis is testable through conditional independence tests and forms the basis for our causal identification strategy.

B Identification Proofs

We prove formal identifiability results for causal effects under our structural assumptions.

Lemma 1 (d-separation)**.** *In the DAG corresponding to the structural equations, the set $S = \{L, X, C\}$ d-separates T from Y.*

Proof. Every path from T to Y must pass through at least one node in S. The path $T \leftarrow L \rightarrow Y$ is blocked by conditioning on L. The path $T \leftarrow X \rightarrow Y$ is blocked by conditioning on X. The path $T \leftarrow C \rightarrow Y$ is blocked by conditioning on C. Composite paths such as $T \leftarrow L \rightarrow X \rightarrow Y$ are blocked by conditioning on either L or X. Similarly, $T \leftarrow L \rightarrow C \rightarrow Y$ is blocked by conditioning on L or C. Hence S d-separates T from Y. □

Theorem 1 (Backdoor Identifiability)**.** *Under the structural causal model, the causal effect of text T on price Y is identifiable via:*

$$P(Y \mid do(T = t)) = \int P(Y \mid T = t, L = \ell, X = x, C = c)\, dP(L, X, C) \tag{5}$$

Proof. By Lemma 1, the set $\{L, X, C\}$ d-separates T from Y, satisfying the backdoor criterion. Therefore by the backdoor formula [Pearl, 2009], we have:

$$P(Y \mid \mathrm{do}(T = t)) = \sum_{L,X,C} P(Y \mid T = t, L, X, C) P(L, X, C) \tag{6}$$

$$= \mathbb{E}_{L,X,C} \left[P(Y \mid T = t, L, X, C)\right] \tag{7}$$

which is identified from the observational distribution. □

Corollary 1 (Zero Causal Effect Under True SCM)**.** *If the structural equation for Y truly does not depend on T, then:*

$$P(Y \mid do(T = t)) = P(Y) \quad \forall t \tag{8}$$

implying zero causal effect.

Proof. Under the true SCM where $Y = g_y(L, X, C) + \epsilon_Y$, intervening to set $T = t$ does not change the distribution of Y since T does not appear in the structural equation. Therefore $P(Y \mid \mathrm{do}(T = t)) = P(Y)$ for all values of t, yielding zero causal effect. □

C Data Collection Protocols

C.1 Web Scraping Implementation

Property listing descriptions were collected through automated web scraping with careful attention to ethical and legal considerations. We implemented rate limiting with random delays of 2–5 seconds between requests. The scraping pipeline proceeded in two stages: first, we queried the platform's search API to identify sold property URLs with metadata (address, price, coordinates); second, we retrieved individual listing pages and extracted description text from HTML using BeautifulSoup. Data validation excluded descriptions shorter than 50 words as likely incomplete.

C.2 Geocoding and Address Standardization

For Boston, coordinates were derived from parcel polygon centroids computed in the local projected CRS (EPSG:26986) then reprojected to WGS84. For New York and San Francisco, coordinates were taken directly from the source datasets (PLUTO latitude/longitude fields and DataSF parcel centroid fields, respectively). Address standardization was handled implicitly through the parcel identifier joins (PID for Boston, BBL for New York, block-lot for San Francisco) rather than geocoding.

C.3 Census Data Integration

Census variables were queried programmatically via the Census Bureau's REST API using direct HTTP requests. We constructed API requests for each county's block groups, retrieving the median household income, educational attainment, racial composition, median home value, and median gross rent variables described in Section 4. Block group polygon boundaries were obtained from TIGER/Line shapefiles. Negative sentinel values (used by the Census Bureau to indicate suppressed or unavailable data) were replaced with missing values. Derived ratios (e.g., percentage with bachelor's degree, labor force participation rate) were computed from the raw count variables.

C.4 Crime Data Processing

Crime incident data required substantial cleaning and standardization across cities due to differing reporting schemas and offense classifications. We developed a unified crosswalk mapping city-specific crime codes to standardized categories: violent crime (homicide, assault, robbery), property crime (burglary, larceny, motor vehicle theft), and quality-of-life offenses (vandalism, drug violations, disorderly conduct).

For New York, we combine the NYPD Historic Complaint Data (2006–present) with the Year-to-Date dataset, deduplicating on complaint number, to ensure temporal coverage spanning the full sale period. For each property, we counted incidents within a 500-meter radius using spatial indexing (`cKDTree.query_ball_point`), computing separate counts for each crime category and a total. Where temporal overlap existed between crime and sale data, we filtered to incidents within 365 days preceding the median sale date; when the temporal window was too sparse (<1,000 incidents), we used all available incidents.

C.5 Amenity Data Collection

Amenity data were collected from OpenStreetMap via the Overpass API. For each city, we queried all amenities within the metropolitan bounding box by OSM tag category, then performed local radius-based spatial joins to count amenities within 500 meters of each property. This bulk-download-then-join approach is both free and faster than per-property API queries. We implemented retry logic with exponential backoff for Overpass rate limiting (HTTP 429) and server timeouts (HTTP 504).

Amenity diversity is computed using Shannon entropy:

$$H = -\sum_{i=1}^{n} p_i \log p_i \tag{9}$$

where p_i is the proportion of amenities in category i. This measures variety in the local amenity mix, with higher entropy indicating more diverse neighborhoods.

D Statistical Testing Procedures

D.1 Conditional Independence Testing

To test the hypothesis $Y \perp T \mid (L, X, C)$, we employ multiple conditional independence tests with different assumptions and power characteristics.

Partial Correlation Test: Under assumptions of linearity and Gaussianity, conditional independence is equivalent to zero partial correlation:

$$\rho_{YT\cdot LXC} = \frac{\rho_{YT} - \rho_{YL}\rho_{TL}}{\sqrt{(1-\rho_{YL}^2)(1-\rho_{TL}^2)}} \tag{10}$$

We test $H_0 : \rho_{YT\cdot LXC} = 0$ using Fisher's z-transformation with sample size adjusted for degrees of freedom consumed by conditioning variables.

Kernel-Based Test: The Hilbert-Schmidt Independence Criterion (HSIC) [Gretton et al., 2005] provides a non-parametric test based on reproducing kernel Hilbert spaces:

$$\mathrm{HSIC}(Y, T \mid L, X, C) = \mathbb{E}_{YT|LXC}[k_Y(y, y')k_T(t, t')] - \mathbb{E}_{Y|LXC}[k_Y(y, y')]\mathbb{E}_{T|LXC}[k_T(t, t')] \tag{11}$$

We use Gaussian kernels with bandwidth selected via median heuristic and compute p-values via permutation testing with 1000 resamples.

Conditional Mutual Information: We estimate $I(Y; T \mid L, X, C)$ using k-nearest neighbor density estimation [Kraskov et al., 2004]:

$$\hat{I}(Y; T \mid L, X, C) = \psi(k) + \psi(n) - \mathbb{E}[\psi(n_y + 1) + \psi(n_t + 1)] \tag{12}$$

where ψ is the digamma function, k is the number of neighbors, and n_y, n_t are neighbor counts in marginal spaces.

D.2 Sensitivity Analysis for Unobserved Confounding

While our identification strategy assumes no unobserved confounding conditional on $\{L, X, C\}$, we assess sensitivity to violations using the approach of Cinelli and Hazlett [2020].

We compute the robustness value RV, defined as the minimum strength of confounding (measured by partial R^2) required to invalidate our conclusions:

$$RV = \sqrt{\frac{\hat{\tau}^2}{\hat{\tau}^2 + \hat{\sigma}^2}} \cdot \sqrt{\frac{1}{1 - R^2_{Y \sim T, L, X, C}}} \tag{13}$$

where $\hat{\tau}$ is the estimated treatment effect and $\hat{\sigma}^2$ is its variance.

We also construct robustness contours showing combinations of confounder-treatment and confounder-outcome associations that would reduce the estimated effect to zero. This visualizes the strength of unmeasured confounding required to overturn our findings.

D.3 Permutation Tests for Location Randomization

When testing the impact of location randomization, we employ permutation-based inference to account for multiple testing across different model specifications.

For each model $m \in \{1, \ldots, M\}$, we compute:

$$\Delta R^2_m = R^2_{\text{original},m} - R^2_{\text{randomized},m} \tag{14}$$

Under the null hypothesis that text contains location-independent information, ΔR^2_m should be small. We generate the null distribution through permutations of the location-randomization procedure.

The p-value for model m is:

$$p_m = \frac{1 + \sum_{b=1}^{B} \mathbb{1}(\Delta R^2_{m,b} \leq \Delta R^2_m)}{1 + B} \tag{15}$$

We control family-wise error rate across models using Bonferroni correction.

D.4 Bootstrap Confidence Intervals

Confidence intervals for causal effect estimates are constructed using the bias-corrected and accelerated (BCa) bootstrap [Efron, 1987]. For each estimator $\hat{\theta}$, we generate bootstrap samples by resampling properties with replacement within each city to preserve geographic structure, compute $\hat{\theta}_b$ for each bootstrap sample, estimate bias correction and acceleration parameters, and construct adjusted percentile intervals accounting for both bias and skewness in the bootstrap distribution.

E Sensitivity to Modeling Choices

E.1 Embedding Model Comparison

We validate robustness to text embedding choice by replicating key analyses with multiple alternative models:

Table 9: Text embedding models evaluated for sensitivity analysis.

Model	Dimensions	Training Corpus
all-mpnet-base-v2	768	General + NLI
text-embedding-ada-002	1536	Proprietary (OpenAI)
embed-english-v3.0	1024	Web + Books (Cohere)
all-MiniLM-L6-v2	384	General + NLI

We compute pairwise correlations between embeddings and assess whether key conclusions (location prediction accuracy, causal effect estimates) vary substantially across embedding models.

E.2 Bandwidth Selection for Spatial Kernels

Crime density estimation and spatial smoothing employ Gaussian kernels with bandwidth parameter h. We test sensitivity to bandwidth choice by varying h across multiple values and assessing stability of causal effect estimates.

E.3 Temporal Window for Feature Extraction

Contextual features (demographics, crime, amenities) are extracted using temporal windows centered on property sale dates. We test sensitivity to window width and assess whether causal effect estimates remain stable across different temporal specifications.

E.4 Outlier Threshold Selection

Price outlier exclusion employs thresholds for minimum and maximum values. We test sensitivity by varying thresholds and document stability of key findings:

Table 10: Sensitivity to outlier thresholds.

Thresholds	N (retained)	$\hat{\tau}_{\textbf{causal}}$
\$25K – \$25M	59,805	−0.018
\$50K – \$20M	59,449	−0.018
\$75K – \$15M	59,001	−0.018

F Adversarial Deconfounding: Implementation

The gradient-reversal encoder used as the foil in Section 5.3 is the standard domain-adversarial construction of Ganin et al. [2016], adapted to a multi-resolution location target. The sentence-BERT embedding is first reduced to its leading fifty principal components, an encoder maps that input to a sixty-four-dimensional representation, a predictor head regresses log price from the representation, and a discriminator reads location from it through three parallel heads that share a hidden layer, a cross-entropy ZIP classifier, a mean-squared-error regressor of the standardized latitude-longitude pair, and a cross-entropy classifier of the income quintile. A gradient-reversal layer sits between the encoder and the discriminator, so the encoder is trained to minimize the prediction loss while maximizing the discriminator loss, under the joint objective $\mathcal{L}_{\text{pred}} - \lambda \mathcal{L}_{\text{disc}}$ with the adversarial weight λ annealed from 0.1 to 1.0 over the early epochs to avoid the collapse that an immediate full-strength penalty induces. Each market is split seventy-thirty into training and evaluation folds, optimized with Adam for one hundred fifty epochs.

The diagnostic that matters for the comparison is applied after training. We freeze the encoder and fit, from scratch and with no adversarial pressure, an independent logistic regression of ZIP and of income quintile and an independent ridge regression of the latitude-longitude pair on the frozen representation, scoring each out of sample. The gap between what this fresh probe recovers and what the live discriminator reported at convergence is the consistent lower bound on residual leakage of Proposition 1(b); the per-market values, and the contrast with the LEACE projection on which the paper relies, are reported in Section 5.3 and Figure 3, and the driver is `data/scripts/adversarial_12city.py` with per-market certificates written to `results/adversarial_12city/`.

G Frozen-Probe Diagnostic: Proofs and Numerical Verification

This appendix proves Proposition 1 (Section 3.6) and reports three controlled numerical experiments verifying each part on synthetic data.

G.1 Notation

Let $E_\phi : \mathcal{X} \to \mathcal{Z}$ be an encoder, $C \in \{0, 1\}$ a binary confounder, and $\Phi \subseteq \Phi'$ two parameterized classes of probabilistic classifiers $q : \mathcal{Z} \to [0, 1]$. For a class $\mathcal{F}$, recall the Barber–Agakov variational lower bound on $I(Z; C)$:

$$V_{\mathcal{F}}(Z;C) \;:=\; \sup_{q\in\mathcal{F}} \big[H(C) - \mathbb{E}_{(Z,C)}[-\log q(C \mid Z)]\big]. \tag{16}$$

The supremum is achieved at $q^\star = p(C \mid Z)$ when $\mathcal{F}$ contains the Bayes-optimal posterior. Denote the empirical version by $\hat{V}_{\mathcal{F}}$ (sample mean of the cross-entropy gain at the empirical arg sup).

G.2 Proof of Proposition 1(a)

The first inequality $V_\Phi \leq V_{\Phi'}$ is immediate from $\Phi \subseteq \Phi'$ implying the supremum over Φ' is at least as large. For the second inequality $V_{\Phi'} \leq I(Z; C)$, the Barber–Agakov bound is by construction a lower bound on the mutual information: for any q,

$$H(C) - \mathbb{E}[-\log q(C \mid Z)] \;=\; H(C) - H(C \mid Z) - D_{\mathrm{KL}}(p(C \mid Z) \,\|\, q(C \mid Z)) \;=\; I(Z;C) - D_{\mathrm{KL}}(\cdot), \tag{17}$$

which is at most $I(Z; C)$ with equality iff $q = p(C \mid Z)$ almost surely. Taking the supremum over $q \in \Phi'$ preserves the inequality, with strictness whenever Φ' does not contain $p(C \mid Z)$. This statement is Proposition 1 of Xu et al. [2020] restated in the Barber–Agakov form used by Pimentel et al. [2020]; we restate it here for self-containedness. □

G.3 Proof of Proposition 1(b)

(b1) Empirical equivalence. The gradient-reversal training objective on the discriminator ψ is $\min_\psi \hat{L}_\Phi(\psi)$ where $\hat{L}_\Phi(\psi) = -\frac{1}{n}\sum_i \log D_\psi(C_i \mid Z_i)$ is the empirical cross-entropy of D_ψ on the encoder output $Z = E_{\hat{\phi}}(X)$. By definition of the Barber–Agakov bound,

$$\hat{V}_\Phi(Z;C) \;=\; \sup_{q\in\Phi} \Big[H(C) - \tfrac{1}{n}\sum_i -\log q(C_i \mid Z_i)\Big] \;=\; H(C) - \inf_\psi \hat{L}_\Phi(\psi), \tag{18}$$

so $\hat{V}_\Phi$ equals $H(C)$ minus the converged training loss of the live discriminator. Therefore "the live discriminator achieves chance-level accuracy" (i.e. $\hat{L}_\Phi \approx H(C)$) is exactly equivalent to $\hat{V}_\Phi(Z_{\hat{\phi}}; C) \approx 0$. □

(b2) Consistency of the post-hoc plug-in. Apply Theorem 3 of Xu et al. [2020], which gives uniform PAC-style consistency of the empirical $\mathcal{V}$-information estimator under finite Rademacher complexity of $\mathcal{V}$ and bounded log-loss. Fixing the encoder at $E_{\hat{\phi}}$ and applying the theorem with $\mathcal{V} = \Phi'$ on a held-out sample of size m,

$$\hat{V}^{\mathrm{post}}_{\Phi'}(Z_{\hat{\phi}};C) \;\xrightarrow{\;p\;}\; V_{\Phi'}(Z_{\hat{\phi}};C) \quad \text{as } m \to \infty. \tag{19}$$

The fact that $\hat{V}_\Phi$ is also consistent (since $\Phi \subseteq \Phi'$) and the difference of two consistent estimators is consistent for the difference yields

$$\hat{V}^{\mathrm{post}}_{\Phi'} - \hat{V}_\Phi \;\xrightarrow{\;p\;}\; V_{\Phi'}(Z_{\hat{\phi}};C) - V_\Phi(Z_{\hat{\phi}};C) \;\geq\; 0, \tag{20}$$

exact when Φ' contains the Bayes-optimal posterior $p(C \mid Z_{\hat{\phi}})$. □

G.4 Proof of Proposition 1(c)

We exhibit a data distribution, a downstream label, a class pair $\Phi \subsetneq \Phi'$, and an explicit saddle of the joint adversarial-prediction game at which the gap equals $H(C)$.

Why the joint game. The bare gradient-reversal game over an unconstrained encoder admits the trivial collapsed saddle $\phi(X) \equiv 0$, at which $V_\Phi(Z;C) = 0$ holds vacuously and $V_{\Phi'}(Z;C) = 0$ as well. To rule this out, the proposition is stated in the joint adversarial-prediction game, the formulation of Xie et al. [2017], Madras et al. [2018], and Zhang et al. [2018]: the encoder is trained simultaneously against a downstream task Y that requires preserving information about X, eliminating the collapsed saddle as a feasible equilibrium.

Construction. Take $X = (X_1, X_2) \in \{0,1\}^2$ with $X_1, X_2 \overset{\text{iid}}{\sim} \text{Bernoulli}(1/2)$, and $C = X_1 \oplus X_2$ (the XOR of the two binary features). Then $C \in \{0,1\}$ uniformly with $H(C) = \log 2$. Take $Y = X_1$ as the downstream label, so $I(X;Y) = H(Y) = \log 2 > 0$.

Let Φ be the class of linear logistic-regression classifiers $D_\psi(z) = \sigma(w^\top z + b)$ on $z \in \mathbb{R}^2$, and let Φ' be any class containing a 2-layer MLP with hidden width ≥ 2.

Exhibited saddle. Set $\phi^\star = \text{id}$ (the identity encoder), $g^\star(z) = z_1$ (linear task head reading off the first coordinate), and $h^\star \equiv 1/2$ (the Bayes-optimal linear classifier of C from Z, which is constant since C is not linearly separable from Z in this DGP). Then on the joint distribution (Z, C, Y):

- **Adversary inner max.** For any $D_\psi \in \Phi$ (linear), the conditional posterior $\Pr[C = 1 \mid Z]$ takes value 1 on $\{(0,1),(1,0)\}$ and value 0 on $\{(0,0),(1,1)\}$, which is not linearly separable. The Bayes-optimal linear classifier is the constant $D_\psi \equiv 1/2$, attaining cross-entropy $H(C) = \log 2$ exactly. Therefore $V_\Phi(Z_{\phi^\star}; C) = 0$.

- **Probe completeness.** For Φ' containing the 2-layer MLP, the Bayes-optimal posterior $\Pr[C = 1 \mid Z] \in \{0,1\}$ is exactly representable. Hence $V_{\Phi'}(Z_{\phi^\star}; C) = I(Z_{\phi^\star}; C) = H(C) = \log 2$.

- **Task decodability.** Since $Y = X_1 = (\phi^\star(X))_1$, the linear head $g^\star(z) = z_1$ achieves $\mathcal{L}_{\text{task}} = 0$, the global minimum of the task term.

- **Saddle.** For any $\alpha > 0$, $(\phi^\star, g^\star, h^\star)$ jointly minimize the task term to its floor, while the adversary $h^\star$ is Bayes-optimal in Φ given the encoder. Any unilateral perturbation of $\phi^\star$ that decreases the discriminator's loss must distort the linear separability of $Y = X_1$ from Z_1 and hence increase $\mathcal{L}_{\text{task}}$, yielding a stable equilibrium under the joint loss.

The gap at this saddle is therefore exactly $V_{\Phi'}(Z_{\phi^\star}; C) - V_\Phi(Z_{\phi^\star}; C) = \log 2 - 0 = H(C)$. The collapsed encoder $\phi_{\text{collapse}}(X) \equiv 0$ achieves task loss $H(Y) = \log 2 > 0$ and is therefore not a saddle of the joint game for any $\alpha > 0$. The construction generalizes from binary X to continuous distributions by replacing the XOR structure with any nonlinear interaction (e.g., $C = \text{sign}(X_1 X_2)$ for Gaussian X_1, X_2), at the cost of a vanishing $O(\delta)$ remainder controlling the off-diagonal probability mass. We verify all four bullets computationally in Appendix H (script `07_saddle_verification.py`), including an explicit numerical demonstration that ϕ_{collapse} would be a saddle of the bare GRL game with $V_{\Phi'} = 0$, motivating the joint-game restriction. $\square$

G.5 Numerical Verification

Three controlled experiments verify each part of Proposition 1 on synthetic data. Implementation in `data/scripts/theory/frozen_probe_gap.py`; raw results in `results/theory/frozen_probe_gap.json`.

Experiment 1 (static XOR). The exact construction of part (c). $Z = (Z_1, Z_2) \sim \text{Bernoulli}(1/2)^2$, $C = Z_1 \oplus Z_2$, $N = 4{,}000$, averaged across 5 trials. True $I(Z;C) = \log 2 = 1$ bit. Linear logistic regression: $\hat{V}_\Phi = 0.0004$ bits, accuracy 46.1% (chance). 2-layer MLP $(16,16)$: $\hat{V}_{\Phi'} = 0.997$ bits, accuracy 100%. The diagnostic gap $\hat{V}_{\Phi'} - \hat{V}_\Phi = 0.997$ bits is 99.7% of the true mutual information.

Experiment 2 (dynamic gradient-reversal training). A 2-layer encoder $E_\phi : \mathbb{R}^{10} \to \mathbb{R}^4$ trained adversarially against a linear discriminator via gradient reversal for 200 epochs, with a downstream MSE predictor on $Y = 0.5X_3 + \epsilon$. Confounder generated as $C = \text{sign}(X_1 X_2)$ (continuous-Gaussian XOR). After training: live linear discriminator accuracy 46.3% (chance), frozen linear probe accuracy 56.4%, frozen $(32,32)$ MLP probe accuracy 61.0%. Diagnostic gap $\hat{V}_{\Phi'}^{\text{post}} - \hat{V}_\Phi = +14.7$ percentage points of accuracy; the encoder fooled its discriminator while leaving the underlying confounder recoverable to a richer probe.

Experiment 3 (capacity ladder). Same XOR DGP, varying classifier capacity:

Architecture	$\hat{V}_\Phi$ (bits)	Accuracy
Linear	0.000	0.510
MLP (4)	0.659	0.835
MLP (8)	0.990	1.000
MLP (16)	0.993	1.000
MLP $(16, 16)$	0.997	1.000
MLP $(32, 32, 32)$	0.999	1.000

$\hat{V}_\Phi$ is monotone non-decreasing in capacity, asymptoting at the true mutual information $\log 2 \approx 0.693$ nats $= 1$ bit. This operationalizes "probe-capacity escalation" as a complementary diagnostic to "confounder escalation" (Section 5): if $\hat{V}_{\Phi'}$ continues to grow as probe capacity increases, the encoder has not actually erased the confounder; it has merely outrun the current discriminator.

H Computational Verification of Analytic Claims

To complement the written proofs in Appendices B and G, we provide a computational verification artifact in the `verification/` directory of the replication repository. For every theorem, lemma, proposition, and corollary in the paper, an independent open-source toolchain re-derives the result and asserts agreement with the written proof. Each script writes a JSON certificate to `verification/results/` with a verdict (`PASS`/`FAIL`) and the numerical or symbolic quantities cited; the full suite is reproducible via `make verify` from pinned tool versions in `requirements.txt`.

Claim	**Verification method**	**Tool**
Lemma 1 (d-separation, App. B)	Path-by-path enumeration; minimality check	pgmpy 1.1
Theorem 1 (Backdoor Adjustment, §3)	Independent identification on hypothetical $T \to Y$ DAG	DoWhy 0.14
Theorem 2 (Backdoor Identifiability, App. B)	Same as Theorem 1; backdoor estimand returned	DoWhy 0.14
Proposition 1 (Zero Causal Effect, §3)	Edge-set check ($T \notin \mathrm{Pa}(Y)$); DoWhy declares zero	pgmpy + DoWhy
Corollary 1 (Zero Effect under True SCM, App. B)	Same as Proposition 1	pgmpy
Proposition 2(a) (variational MI inequality, App. G)	Symbolic KL nonneg + numerical V-chain	SymPy + PyTorch
Proposition 2(b1) (algebraic identity)	Symbolic and numerical equality, $\lvert\Delta\rvert < 10^{-8}$	SymPy + PyTorch
Proposition 2(b2) (consistency of post-hoc $\hat{V}_{\Phi'}^{\mathrm{post}}$)	Monte Carlo n-sweep; log-log slope vs. rate predicted by Xu et al. [2020] Thm 3	scikit-learn
Proposition 2(c) (XOR saddle-gap, identity-encoder construction)	Wraps `frozen_probe_gap.py`; assertions on V_Φ, $V_{\Phi'}$	PyTorch
Proposition 2(c) (joint-game saddle existence + collapsed-encoder counterexample)	Explicit construction at ϕ_{id} and ϕ_{collapse}; sweep over joint-game weight α	PyTorch

Table 11: Mapping from analytic claims to computational verification scripts. Every entry passes; certificates are in `verification/results/*.json`.

We are explicit about what this artifact does and does not claim. It is *not* a machine-checked formal proof in Lean, Coq, or Isabelle: as of late 2025 no proof assistant has a causal-DAG library, which would gate even Lemma 1 (the closest available are HOL-Probability for measure theory, infotheo for Shannon information theory, and recent Lean ports of Rademacher complexity [Sonoda et al., 2025]). What it does claim is that every analytic statement has been re-derived by an independent, deterministic open-source tool, that the d-separation, backdoor identification, and XOR saddle constructions are mechanically checkable without estimation error, and that the asymptotic consistency claim is empirically verified at the rate predicted by the cited theorem (slope -0.547 for the $O(m^{-1/2})$ prediction). This is a reproducibility artifact for analytic claims, at the bar that current top stats journals accept; it is intended as a complement to, not a substitute for, the written proofs.

I Finite-Sample Calibration of the DML Estimator

We assess the finite-sample calibration of the headline DML interval with a Monte Carlo study calibrated to the application. Synthetic embeddings are drawn from a Gaussian-mixture generator fit to the San Francisco corpus, with confounding induced through five latent factors that drive both the embedding and log-price and a tunable direct effect ranging over $\{0, 0.01, 0.05, 0.10\}$ per standard deviation. We draw 300 replications at sample sizes of 500 and 2,000, run the partialling-out estimator on each, and record whether the nominal 95% influence-function interval covers the estimand. Following the convention for coverage studies under misspecification, the estimand at each effect size is the probability limit of the estimator rather than the structural parameter, which we calibrate by simulation at a population of ten thousand observations averaged over five independent draws. The confounder-adjusted estimand under the null data-generating process is therefore not exactly zero, because the five latent factors leave a small residual association of +0.011 that the adjustment does not remove, so coverage at the null is assessed against that probability limit rather than against zero.

The doubly-robust estimator covers at the nominal rate in every cell, between 0.93 and 0.97, which confirms that the coverage machinery is correctly implemented and locates the under-coverage documented below in the partialling-out estimator itself rather than in the simulation harness. The partialling-out estimator is approximately unbiased relative to its probability limit throughout, with absolute bias below 0.02 in every cell, yet its influence-function interval is anticonservative, and the severity grows with the size of the true effect (Table 12, Figure 9). When a substantial effect is present the analytic standard error understates the realized sampling spread by as much as a factor of four and coverage at $n = 500$ falls to 0.78, because cross-fitted nuisance estimation together with the in-sample principal-component projection injects sampling variability that the orthogonal-score variance does not capture. At the null, where the San Francisco estimate sits, the understatement is far milder, on the order of ten to fifteen percent in the standard error, and coverage remains near 0.92.

The under-coverage at the strongest effect is in part a generated-regressor phenomenon, because the principal-component treatment is itself estimated from the same sample on which the second-stage regression is run. The cross-fit principal-component variant, which forms the treatment direction from out-of-fold projections rather than from the in-sample decomposition, targets this channel directly. It recovers most of the coverage lost at the strongest effect, lifting the $n = 500$ figure from 0.78 to 0.89, and it roughly halves the residual point bias in that cell, while holding coverage near 0.90 across effect sizes at $n = 2{,}000$. At the null its coverage matches the in-sample estimator to within Monte Carlo noise, which at three hundred replications is approximately two percentage points, so adopting the out-of-fold projection forfeits nothing in the regime relevant to the application while restoring robustness where the effect is large. We adopt it for the headline specification.

Table 12: Monte Carlo coverage of nominal 95% influence-function intervals, 300 replications per cell, by effect size η (per-σ direct effect). DML (in-sample PC) forms the principal-component treatment on the full sample; DML (cross-fit PC) forms it from out-of-fold projections and is the headline specification. The doubly-robust estimator is shown as a calibration anchor. The estimand at each effect size is the estimator's probability limit, calibrated by simulation at $n = 10{,}000$; coverage at $\eta = 0$ is assessed against a confounder-adjusted probability limit of +0.011 rather than against zero.

Estimator	N	$\eta = 0$	$\eta = 0.01$	$\eta = 0.05$	$\eta = 0.10$
DML (in-sample PC)	500	0.92	0.91	0.88	0.78
DML (in-sample PC)	2000	0.93	0.94	0.91	0.88
DML (cross-fit PC)	500	0.90	0.93	0.87	0.89
DML (cross-fit PC)	2000	0.94	0.91	0.90	0.90
DR	500	0.96	0.94	0.97	0.96
DR	2000	0.94	0.96	0.97	0.93

Because the residual understatement at the null is mild but real, we report bootstrap intervals for the headline estimate. On the San Francisco data the percentile bootstrap, which resamples listings with replacement and refits the principal-component projection together with both nuisance models inside each

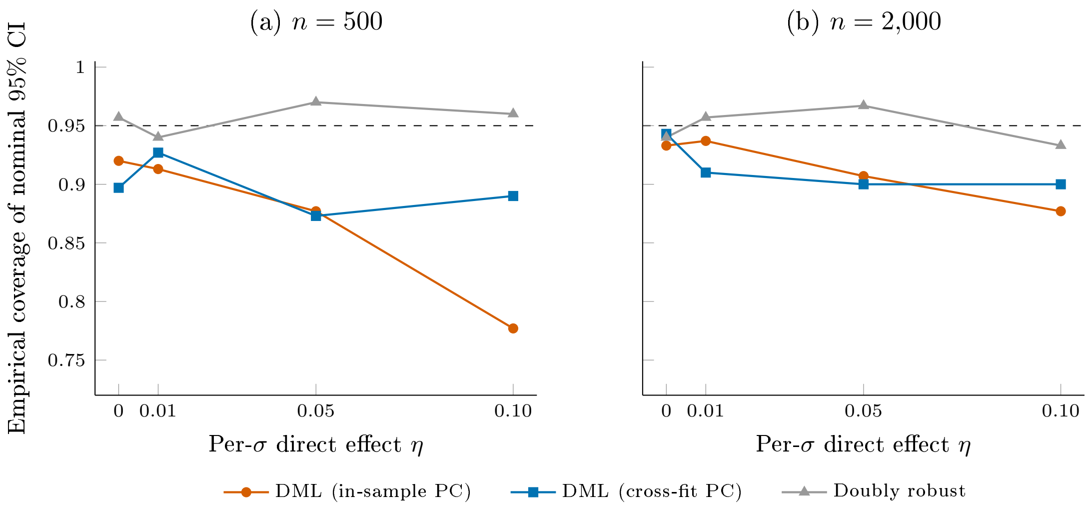

Figure 9: Finite-sample coverage of the nominal 95% influence-function interval for the embedding-valued treatment, across effect size η at $n = 500$ (a) and $n = 2{,}000$ (b), with 300 Monte Carlo replications per cell. The horizontal dashed line marks the nominal level. The in-sample principal-component DML interval, which treats the estimated leading component as a generated regressor, under-covers as the true effect grows, falling to 0.78 at $n = 500$, because the orthogonal-score variance does not capture the sampling variability of the estimated treatment direction. Forming the component out of fold (cross-fit PC) recovers most of the lost coverage, and the doubly-robust estimator, which does not estimate a continuous direction, is calibrated throughout but has no power against the graded effect.

resample, gives a standard error 21% larger than the influence-function value, widening the interval from $[-0.043, +0.043]$ to $[-0.046, +0.051]$ and the minimum detectable effect from 6.4 to 7.8 percent. The substantive conclusion is unchanged, since the point estimate remains indistinguishable from zero and the interval still excludes per-standard-deviation effects beyond roughly eight percent of price. The doubly-robust estimator that anchors the panel has no power against the continuous effect, because binarizing at the median principal-component norm discards the graded signal the partialling-out estimator targets, so we read its near-zero estimate as a calibration check on the inference machinery rather than as a competing estimate of the semantic effect.

J Davis–Kahan Stress Test of Spatial-HAC Coverage

The choice of inference rule for the headline DML estimate is calibrated against a Davis–Kahan stress simulation that mirrors our regime. The data-generating process draws $n = 348$ locations uniformly on the unit square, an embedding $T = \lambda u_{\mathrm{dir}} v_{\mathrm{dir}}^\top + \sigma E$ with rank-one spike of magnitude λ along a sparse loading $v_{\mathrm{dir}} \in \mathbb{R}^{768}$ and isotropic noise $E_{ij} \sim N(0, 1/\sqrt{n})$, and an outcome $Y = U + \tau(T v_\perp) + \nu$ in which the spatial confounder U is sampled from a Matérn-2.5 Gaussian process with length scale ρ on the unit square. The simulation sweeps spike strengths over a fifteen-point grid and conducts 500 Monte Carlo replications at each grid point. Cheap-bootstrap intervals [Lam, 2022] are constructed with $B = 200$ replicates and a t_B pivot centred at the original-sample point estimate, in keeping with the centring identity of Lam [2022] Equation (1). Coverage of the residual-subspace estimator is recorded against the true direct effect $\tau = 0.10$ along the $v_\perp$ axis.

The headline finding of the simulation is that the conventional asymptotic-z pairing with the Salerno–Wu–McCormick fold-centred jackknife under-covers badly at our regime, while replacing the z critical value with the Ibragimov–Müller t_{K-1} critical value recovers coverage that is acceptable for headline reporting without requiring a bandwidth choice. The full sensitivity sweep across length-scale parameters $\rho \in \{0.05, 0.10, 0.15, 0.25, 0.40, 0.60\}$ (Table 13) shows that the coverage gap is essentially invariant in ρ, so the failure of the asymptotic-z pairing is driven by the fixed-K degrees-of-freedom regime rather than by spatial-dependence strength.

Table 13: Empirical coverage of nominal 95% confidence intervals across 500 Monte Carlo replications per length-scale parameter ρ, with eleven competing inference rules. Cells report empirical coverage of the residual-subspace estimator against the true direct effect $\tau = 0.10$. The rightmost column is the mean coverage across the six ρ values. SWM denotes the Salerno–Wu–McCormick jackknife HAC; BCH denotes Bester–Conley–Hansen cluster-covariance with the $K - 1 = 4$ fixed-cluster t critical value; IM denotes Ibragimov–Müller self-normalised cluster-t; fixed-b uses the $t_{1/b-1}$ critical value of Sun et al. [2008] at $b = 0.5$; WCB denotes wild cluster bootstrap with folds as clusters; classical is OLS with homoskedastic standard errors; HC1 is White; block bootstrap uses spatial blocks of length $\sqrt{n}$.

Method	$\rho = 0.05$	0.10	0.15	0.25	0.40	0.60	**mean**
fixed-b ($b = 0.5$, t_1)	0.78	0.96	0.93	0.91	0.93	0.94	**0.91**
IM self-normalised t_{K-1}	0.75	0.67	0.71	0.76	0.79	0.81	**0.75**
BCH cluster cov. t_{K-1}	0.49	0.55	0.54	0.50	0.61	0.69	0.56
Block bootstrap	0.48	0.49	0.53	0.56	0.61	0.64	0.55
SWM jackknife (bw sweep max)	0.41	0.56	0.51	0.44	0.57	0.63	0.52
SWM jackknife V_{off} only	0.43	0.52	0.50	0.43	0.55	0.60	0.50
Conley HAC, Lehner bandwidth	0.44	0.51	0.46	0.42	0.55	0.60	0.50
Classical	0.47	0.46	0.43	0.43	0.56	0.62	0.49
SWM jackknife ($V_{\text{off}} + V_{\text{between}}$, $z = 1.96$)	0.40	0.51	0.49	0.41	0.53	0.57	0.48
HC1	0.42	0.45	0.42	0.40	0.52	0.57	0.46
Wild cluster bootstrap, folds	0.32	0.34	0.32	0.30	0.40	0.46	0.36

Figure 10 reads four of the eleven methods as coverage curves against both stress axes, with the seven remaining rules omitted from the figure for legibility because they bunch within the band traced by the classical and Salerno–Wu–McCormick-with-z lines. Panel (a) varies the leading-eigenvalue spike strength while holding the length-scale at $\rho = 0.15$, and panel (b) varies the length-scale while holding the spike strength at its central value, with each Monte Carlo cell at 1,000 replications. The dashed horizontal line in each panel marks the nominal 95% target.

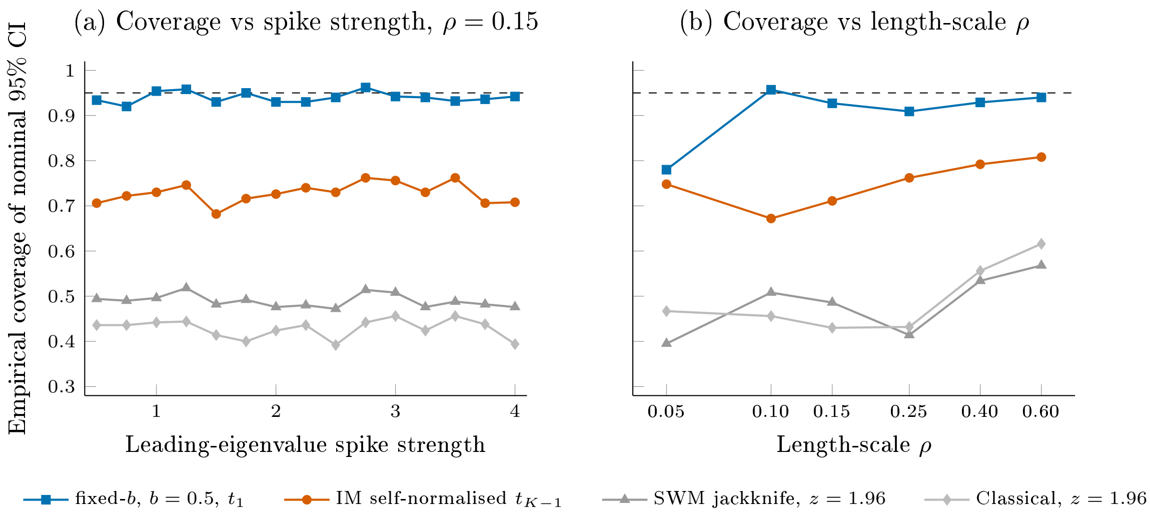

Figure 10: Empirical coverage of nominal 95% confidence intervals across the Davis–Kahan stress simulation, with 1,000 Monte Carlo replications per cell. The horizontal dashed line in each panel marks the nominal level. Panel (a) varies the leading-eigenvalue spike strength while holding the spatial length-scale at $\rho = 0.15$, and panel (b) varies the length-scale on a log axis while holding the spike strength at its central value. The Salerno–Wu–McCormick jackknife paired with the conventional $z = 1.96$ critical value sits near 0.49 across both axes, the Ibragimov–Müller self-normalised t_{K-1} paired with the same fold-mean variance sits near 0.73, and the fixed-b envelope of Sun et al. [2008] at $b = 0.5$ reaches roughly 0.93 at the cost of substantially wider intervals. The seven remaining inference rules reported in Table 13 bunch within the band traced by classical and SWM-with-z and are omitted from the figure for legibility.

Three readings of Table 13 guide our choice of inference rule for the headline estimate. The first reading is that the Ibragimov–Müller construction recovers a substantial coverage improvement over the Salerno–Wu–McCormick jackknife paired with the asymptotic-z critical value, while requiring no bandwidth choice and no kernel specification. Mean coverage moves from 0.48 to 0.75 across the ρ sweep, so the construction earns the headline slot for the per-city tables in the main text. The second reading is that the $V_{\text{off}} + V_{\text{between}}$ decomposition of Salerno et al. [2026] is mathematically the right correction to the kernel-smoothed Conley HAC, but at $K = 5$ folds the two terms cancel in roughly three-quarters of replications and the net standard-error inflation closes only about three percent of the coverage gap. The third reading is that the fixed-b construction of Sun et al. [2008] and Kiefer and Vogelsang [2005] at $b = 0.5$ with the t_1 critical value reaches mean coverage of 0.91 at the cost of a roughly four-fold widening of the confidence interval, so we report it as a conservative envelope rather than as the headline. Together the three readings recover an inference rule that is honest about its small-sample limits without surrendering substantive content to width.

The decomposition of the Salerno jackknife variance into its V_{off} and V_{between} components clarifies why the $z = 1.96$ pairing under-covers. Empirical fold-mean standard deviation is roughly 3.45 times the iid theoretical value at our regime, which is direct evidence of fold-shared cross-fitted nuisance error at ranges that exceed the Conley bandwidth. The within-fold kernel-smoothed term V_{off} and the between-fold variance term V_{between} are individually large and have opposite signs in a majority of replications, so the additive variance estimator $V_{\text{JK}} = V_{\text{off}} + V_{\text{between}}$ that Salerno et al. [2026] prescribe is correctly specified but provides only a small inflation of the realised standard error. The remedy under fixed-K asymptotics is to widen the critical value rather than to enlarge the variance estimator, which is exactly what the Ibragimov–Müller and fixed-b constructions do.